



Gardening Wrist Orthotic

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Executive Summary

Problem Definition Review

1. Introduction

Chronic and temporary joint pain affects a large demographic of people around the world. It stops individuals from participating in hobbies and activities of personal enjoyment and increases their intake of medication for pain relief. All members of Team N were tasked with interviewing an individual about frustrations they may encounter in their daily lives; some of their pains included poor house heating, laptop battery life, lower-back mobility issues, and joint pain while gardening. In this project, Team N has decided to address this problem for individuals who garden, whether professionally or as a hobby, and create a wrist orthotic device to stabilize the wrist and remain adaptable for gardening purposes.

The team has researched the demographic most affected by joint pain, the main contributors to such pain especially in the upper limbs, and alternative products on the market that are targeted toward solving the team's chosen problem. The team created a graphical user persona to mimic the target audience of the project's final product to solidify the intended users, and therefore also determine the most significant factors the team must take into consideration during the product brainstorming process. The team then compared alternative products to their own idea using informational graphics and charts and determined the user needs, factors, and costs required to create this product and ensure its effectiveness surpasses those of the alternatives.

During the brainstorming process, the team created three concept designs for the final product. The team considered the most effective methods of designing and creating the product, taking items like materials, budget, and user needs as the highest priorities. The team also received feedback from other teams to finalize the concept plan for the orthosis. The wrist orthosis will contain a brace around the forearm, wrist, and palm, and will be customized to each user using modeling software. A gardening glove with flexible support plates through the backhand and fingers will be attached to the brace, allowing users to garden comfortably. A quick release plate will be sewn onto the palm of the glove to correspond with another plate attached to an elastic band that can be put on the handle of hand shovels and small gardening tools. The plates will click and attach to provide the user with a firmer grip upon the tool and subsequently further relief on the wrist and joints.

2. Problem Definition

The problem the team aims to address is the lack of accessible assistive devices that allow individuals with joint pain to garden comfortably and safely. The team is exploring the design of an orthosis or similar product intended to reduce pain and strain during gardening activities. The selected user group includes individuals with upper-body joint pain and mobility limitations, caused by either chronic or temporary illness and injury.

a. User Persona

Figure 1 depicts the team’s user persona, named Riley Thomas, who represents a potential individual within the team’s selected target audience. Riley is a 65-year-old Caucasian American woman who lives with her husband in a suburban neighborhood. She has a college education, is currently retired, and frequently gardens in her front and backyard. She has arthritis that especially affects her wrists, elbows, and shoulders, and she often takes medication for pain relief although the medication may have long term effects on her body.

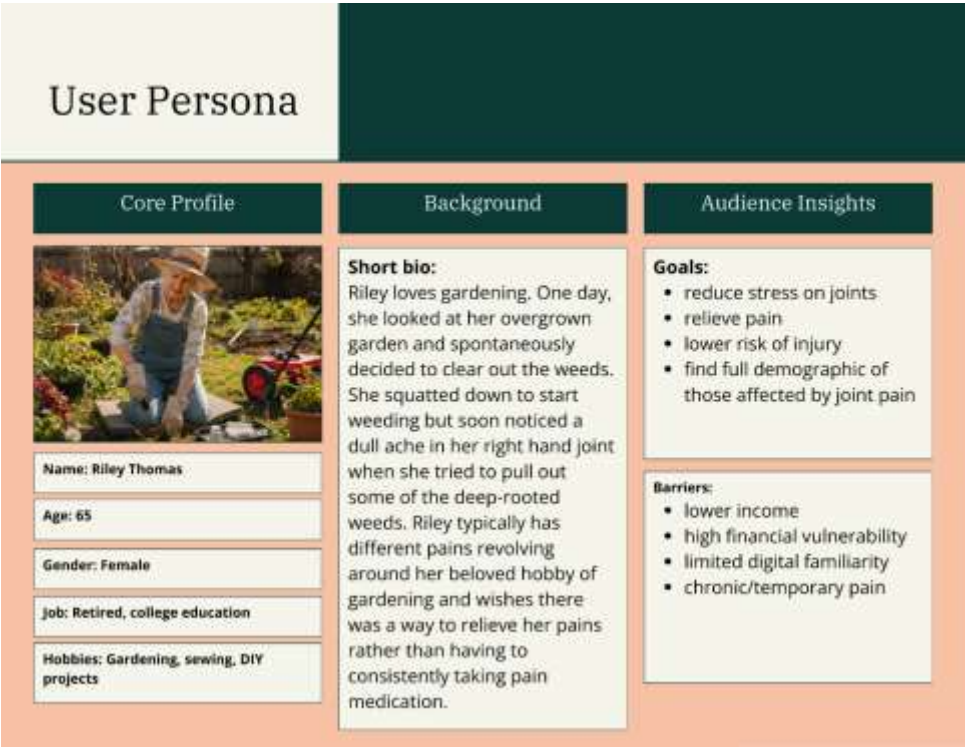


Figure 1. Graphical User Persona

Riley loves gardening. One day, she looked at her overgrown garden and spontaneously decided to clear out the weeds. She squatted down to start weeding but soon noticed a dull ache in her right hand joint when she tried to pull out some of the deep-rooted weeds. Riley typically has different pains revolving around her beloved hobby of gardening and wishes there was a way to relieve her pains.

b. Tasks and Pains

The selected user group has a considerable number of tasks and pains associated with gardening while affected by joint pain. Common gardening tasks include weeding and planting on the ground, forcing individuals to bend and rest on their limbs often. Additionally, digging and turning soil requires the use of heavy tools and repetitive motions that can strain arm joints and may require often taking medicine to relieve pain.

- **Task:** Weeding and planting

- **Pain:** It requires constant, forceful, use of hands and forearms.
- **Pain:** It often requires resting on the arms and elbows.
- **Pain:** It is difficult to reach the roots of the weeds buried in the soil.
- **Pain:** It is often necessary to take many breaks and take pain relief medication.
- **Task:** Digging and turning soil
 - **Pain:** Using shovels can put severe strain on the arms.
 - **Pain:** Can cause tendonitis & carpal tunnel syndrome from gripping tools tightly.
 - **Pain:** Can strain the rotor cuff due to repetitive actions.

3. Research Plan

The goal of conducting research is to gain an understanding of what the most common factors of joint pain and mobility issue plague individuals to have a basis for the anatomical nature of the team's product. Another goal of the team's research is to learn how to design such a product, particularly using three-dimensional modeling software, what materials are needed to build the product, and how the product can be more efficient compared to other similar products in the market. An ethical consideration the team discussed was the environmental concerns that come with using plastic for the basis of the product, and the team's solution to this is using recyclable filament during the 3D printing process.

Table 1. Research Plan Table

Research Question	Qualitative Data Collection	Quantitative Data Collection
(RQ1.1) Which upper-body joints or anatomical parts are largely affected by joint pain? (RQ1.2) What is the demographic of those affected by joint pain?	Online research paper of pain mapping for common shoulder disorders [1].	Online article of a survey study of upper limb pain and disability among university students [2]. Online research article the changing profile of joint disorders with age [3].
(RQ2) What are the common tasks and pains associated with gardening?	Online article and study that characterizes gardening tasks and corresponding body positions done by older adults. [4]	Online research article on hand discomfort among gardeners. [5]

(RQ3) What are orthoses and how do they provide structural support?	Online article covering how different orthotic devices or orthoses are created and used to benefit the users whether they are stabilizing or preventing injury. [6] Online article that lists and explains the different strategies and components in making an orthotic/prosthetic device. [7, 22]	Online article representing the results of a survey where participants were prescribed an orthotic device in which the average score of the participants satisfaction of the device was 4.53/5. [8]
(RQ4) How can 3D modeling be used to create a product?	Online article comparing traditional and additive manufacturing processes and how 3D printing offers potential in developing orthotic devices [9].	Online article showing the effectiveness and quick fabrication process of 3D printed orthotics and prostheses in canine, proving the viability of these products in the medical field [10]. Online research article showing how effectively utilizing CAD and 3D printing technology can manufacture smaller and lighter medical products [11].
(RQ5) What are some other similar products that solve this problem, and how can our product be more efficient?	Online research paper on improving gardening for people with arthritis. [12]	Online conference paper covering orthotic prototypes for patients with osteoarthritis, chronic joint pain. [13]

4. Research Results

a. Creating User Value

Pains

Gardening is undoubtedly a taxing process on your joints, making it harder for some older or injury-prone people to grip tools and have a proper range of motion for the activity. Some orthotic prototypes have already been developed such as an ergonomic chair [12], yet these

solutions are big and unpractical in everyday, gardening use. Researchers analyzed a group of older gardeners, after performing seventeen different common garden tasks, over ten different pains were observed, demonstrating the risk and damage caused by the consistent, scooping, bending, and stretching of the joints and ligaments during gardening [4].

- Using shovels can put severe strain on the arms.
- It can cause tendonitis & carpal tunnel syndrome from gripping tools tightly.
- It can strain the rotor cuff due to repetitive actions.
- Gardening requires constant movement, getting up and down, and frequently resting on elbows
- Can lead to pulled and torn muscles in the arms

Gains

Different 3D printing processes can allow manufacturers and the team to construct different orthotic devices at a quicker pace at a lower cost. It can allow the team to create a model where the team can print 3D parts with different types of filaments that can be changed for different needs. Such as carbon fiber laced filament for durability or TPU for parts that need to be flexible. [9].

- More efficient
 - Will take less time and effort to pull weeds out of the ground
- Lower consumption of pain medication
 - Rely less on pain medication, reducing their spending
- More cost effective (user only needs one tool instead of multiple shovels, etc.)
 - The device will be cheaper than other orthotics on the market
- Adaptive; can change the tools out
 - Users can attach gardening tools that can be interchanged on the device

User Needs

Orthotic devices help the users' needs of reducing pain and having a lower risk of injury by allowing the joints, muscles, and bones to be more structurally sound and stable [6]. The overall goal of this device is to aid and target users with joint pain, ensuring several design requirements/categories are met (cost, pain reduction, etc.). For example, pivoting the orthotic device in a 3D printing direction not only makes production faster, but it is cheaper and more adaptive. Additionally, with recyclable filaments, an eco-friendly approach to orthotics could also be explored [10].

- Performance Efficiency
- Pain Reduction
- Lower Risk of Injury

- Cost-effective
- Adaptive
- Comfortability
- Durability

Pairwise Comparison Chart

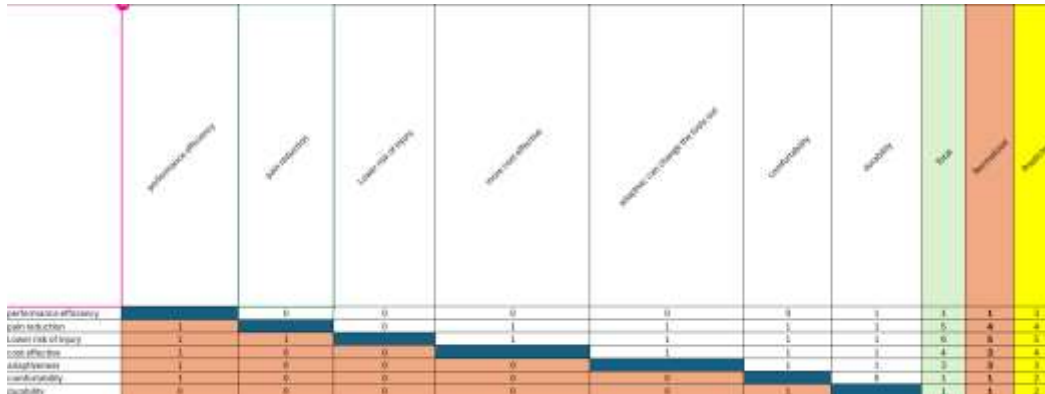


Figure. 2. Pairwise Comparison Chart

With any design process, some users need to become more redundant over time as design focus shifts from idea to idea. As shown by the above comparison chart, needs such as efficiency and durability were only preferred by the team once over other categories, while needs such as pain reduction and injury prevention were more heavily prioritized in the team's brainstorming and development process [14]. While searching for other orthotic devices, the team noticed that most wrist-hand orthotic devices also couldn't balance certain needs when designing, including more commonly cost versus durability and effectiveness. With this as the team's priority, some sacrifices to the final prototype may need to be made, however it is a matter of balancing the needs than homing in on only a few needs [25].

b. Market Character

Stakeholders

- Medical Orthotic Companies
 - These companies could invest in these orthotic products, possibly being able to work with the team to bring these devices to a wider market.
- Urgent Cares/Medical Centers
 - These facilities could buy and distribute them to patients in need when properly consulted by a physician.
- Medical Stores/Pharmacies
 - The team can easily sell and distribute these products nationwide through stores like CVS, Walgreens, etc.

Market Size

The global market for wrist-hand orthotics is already growing at a rapid rate, currently sitting at around 1.8 billion USD in market cap, with a projected 3.1 billion USD market cap by 2033 [14]. Additionally, Wrist Supports Market size was valued at USD 1.5 Billion in 2024 and is forecasted to grow at a CAGR of 7.2% from 2026 to 2033, reaching USD 2.8 Billion by 2033 [15]. With a growing market, there are more target users. The team's device will be used by general wrist-hand orthotic patients but mainly targets users who want to work hands-on, in this case gardening, with their orthotics. The main idea for the team's device is to allow users to complete their daily tasks without unbearable joint pain and with the associated attachable tools, hands-on activities like gardening/weeding are performed with ease.

Current Alternatives

- **Strap Wrench:** Tool that uses a rubber or nylon strap, typically 12–18 inches long, to grip smooth, polished, or irregular surfaces [20]
- **Twist Jar Opener:** A handheld or automatic kitchen tool designed to easily remove jar lids that are stuck or stubborn [19]
- **ACE Carpal Tunnel Wrist Stabilizer:** Wrist brace; “Stabilizer helps relieve symptoms associated with Carpal Tunnel Syndrome with superior, stabilizing support” [18]
- **IMAK Compression Gloves:** Specialized gloves designed to relieve hand pain, stiffness, and swelling associated with arthritis and poor circulation by tightening around the joint [23]
- **PenAgain:** Ergonomic pen; “This wishbone-shaped writing utensil is designed for those with carpal tunnel, arthritis, and anyone with a weak grip or limited grasp. It allows the pressure of the natural weight of your hand, rather than your grip, to guide the writing instrument.” [21]
- **Paexo Wrist:** Mechanical skeleton brace; “Paexo Wrist provides effective support for your wrist when handling loads. The exoskeleton was derived by Ottobock Industrials from a medical device that was specially developed for the treatment of wrist inflammation.” [17]
- **Daiya Wrist Exoskeleton:** “The Daiya wrist exoskeleton increases support to your wrist during load and tools handling. Its thin aluminum support adjusts the level of wrist fixation in stages to reduce pressure, and three pockets containing aluminum supports on the back and palm of the hand allow the user to adjust the level of fixation.” [25]

- **Stainless Steel Tongs:** Multipurpose tongs: “Stainless steel construction: these tongs feature a high quality, 430-stainless steel frame” [24]

User Need	Shop Wrench	Good Grips Twist Air Opener	ACE Brand Carpal Tunnel Wrist Stabilizer	IMAK Compression Arthritis Gloves	RCR Wrist Drop Orthosis	Paragon	Passo Wrist
Performance Efficiency							
Pain Reduction							
Lower Risk of Injury							
Cost Effective							
Adaptiveness							
Comfortability							
Durability							

Figure. 3. Current Alternatives Comparison Chart

Some gaps that the team noticed between the current alternatives are that many of them focus on being cost effective while sacrificing other important aspects such as durability and adaptiveness. The team concluded that the goal would be to try to create a device or product that includes most, if not all, of these attributes to ensure that the product will remain a reasonable price while also allowing it to be durable and adaptive for different uses.

In a survey conducted in 1997, about 30% of the respondents indicated “chronic upper extremity pain associated with physical impairment” [1]. Participants who experienced such pain were a demographic consisting of both men and women of ages 25 to 74 who were randomly selected, and were likely to be blue collar workers, long-term absentees, and current or former absentees. In corroboration with this source, another study conducted in 2019 resulted in a 24% respondent rate of the prevalence of upper limb pain [2]. This study’s subjects were university students between the ages of 18 to 28 where most subjects had been using smartphones for more than 5 years. The results of both studies show that chronic and temporary upper limb pain and discomfort are prevalent across a large demographic, and therefore the team’s decided target audience has not been limited to constricting age, gender, or ethnic range.

Hand tools, such as spades, shovels, sickles, and rakes, are commonly used by gardeners during activities. The excessive use of such tools is often followed by pain or discomfort in gardeners’ hands and forearm regions. Manual work such as gardening requires repetitive movements and can cause high muscular discomfort and strain injuries in the upper limbs due to forearm muscle strain and poor wrist posture. Findings of a 2024 study unveiled that majority of respondents found gardening hand tools to have low to moderate comfort, and there was an increase in pain and discomfort in the hands while using such tools which in turn interfered with their gardening performance [5].

An article published in 2022 showed that traditional orthotics/support devices (such as the spinal cord injury orthotics in this article) are usually made of plaster, and the advent of low-temperature thermoplastic sheets make the finished products bulky, time-consuming (1 week) and expensive [11]. At the same time, due to the limitations of the manufacturing process, their appearance is difficult to make aesthetically pleasing, and their geometric defects also lead to a lack of personalized comfort and functionality. However, by using 3D printing technology, people can design structures that are difficult to make by hand and have extremely high precision. Another study also showed that compared with traditional manufacturing methods, 3D printing technology can produce

an orthotics/support device with personalized customization in terms of fitness and shape in one day, thus providing patients with a more economical, comfortable and personalized product [10]. Additionally, a study focuses on customized orthotics/prostheses also showed that it takes less than 6.5 hours to manufacture a 3D printed personal orthotics using PLA Material on a fused deposition modeling (FDM) machine, and the average operation time is only 10-15 minutes. Compared to traditional methods, the time required to fabricate orthotics for patients using 3D printing technology will be significantly shorter [9].

5. Value Proposition

The team's proposed solution to address the problem of joint pain and mobility issues is designing a supportive brace or glove that can lower the user's pain level. The product is specifically designed for gardening purposes, with the ability to attach tools to the brace that can help users easily weed and plant their gardens. The target end users are individuals over the age of 18 who suffer from temporary or chronic joint pain specifically in the wrists. Since race or gender do not seem to define the demographic of those affected by joint pain [2] but typically adults or older aged individuals find themselves to be more affected [3] [4]. The team has decided to focus on developing the product for those in the age range of 18 and above. Users of this device will experience reduced joint pain and can swap out tools on their device, making strenuous tasks like shoveling or gardening less likely to inflict lasting injuries or pain. Being that this device is 3D printing allows it to be a cost-effective option for users, preventing unbearable joint pain without breaking the bank. With the growing market in these wrist-hand orthotics expecting to double in some regions by 2033 [14], this device will likely be revolutionary, all-in-one for those who want to be active without the worry of serious injury and pain.

6. Design Focus

The team has discussed and concluded that when it comes to the overall design of the brace, the team wants the product to be comfortable for the user. As a result, the team decided that the best options for securing the brace to the user would either be a Velcro type of function to ensure that the product will stay attached to the user comfortably without it pinching too tight or being too loose. Additionally, the team debated on including a feature of convenience such as utilizing a magnetic plate or an area with different straps that allow users to have easy access to different gardening tools without having to often bend to reach the ground or crowd their hands and pockets.

The team decided that the easiest way to design the product is through the Computer Aided Design (CAD) software. The team intends to produce the final product by 3D printing the base structure with filament, as the variety of filaments that can be used can be determined in context of the products need to be more

flexible or durable [11]. These components can also help with the user needs the team identified because filament is environmentally friendly as it can be recycled and additionally, it can typically be stretched out for long distances for a cheaper price, allowing the team to market the product at a more cost-effective standpoint while remaining effective for use.

Concept Development Review

7. Concept Brainstorming and Ideation

a. Process Description

When brainstorming new ideas for the design, the primary method for finding new design options was to research other orthotics, take inspiration from other prototypes and designs, and rework them to fit the gardening orthotic. Another important factor when brainstorming is fitting user needs. When an idea is brought to the team, it is evaluated for how it applies to all the user's needs. Big factors like cost, pain reduction, and injury prevention are focused during this process with other needs being secondary but still vital for the final design. Overall, the main process for the team's concept of brainstorming is to research and compare user needs, finding the most viable and safe option for its users.

b. Brainstorming Results

The team concluded from brainstorming that the final product would focus on pain reduction and adaptiveness and therefore would include a kind of magnet or strap on the palm of the orthosis to attach different gardening tools to the structure. One possible idea discussed for this detail was 3D printing a quick-release clip slot sewn or in some way attached to the palm of the orthosis glove and additionally create a strap with a slotting piece that could be wrapped around any tool. The slotting piece could be inserted into the clip slot on the glove, allowing the user to gain a firm grip on the tool while being able to easily remove the tool. The team discussed methods that could be used to adjust the sizing of the orthosis, such as Velcro, string, or a quick lacing system using turn knobs. The team decided that the lacing system would be rather difficult to implement and deemed Velcro the better solution. For the base structure of the orthosis, the team considered using a fabric compression glove for the palm and fingers and utilizing 3D printing for the wrist brace structure, which could be customized to the user using an online modeling software such as Lidar Scan.

8. Concept Selection

a. Down-Selecting to Two Concepts

Through the team's discussions with other teams, the team was able to pick two final concepts as to what the team's product would include. Many of the other teams liked the idea of a quick-release plate for adaptability as well as the team's idea of how the team was going to secure the device on the user. The two remaining concepts (concept 1 and 2) were selected because both included useful and worthy additions. For the team's final product, students will most likely combine the best aspects from both concepts to achieve the best overall product.

b. Pugh Scoring Matrix

User Needs	Weight (1-5)	Reference (RCAI)		Concept 1		Concept 2	
		Rating	Weighted Score	Rating	Weighted Score	Rating	Weighted Score
Performance Efficiency	1	5	5	4	4	4	4
Pain Reduction	4	4	16	5	20	3	12
Lower Risk of Injury	5	3	15	4	20	5	25
Cost Effective	4	1	4	4	16	3	12
Adaptiveness	3	5	15	3	9	4	12
Comfortability	1	4	4	3	3	3	3
Durability	1	5	5	4	4	3	3
			0		0		0
			0		0		0
			0		0		0
TOTAL			64		76		71

Figure. 4. Pugh Scoring Matrix Chart

These results show that the reference design (RCAI Wrist Orthotic), while being superior in some categories, struggled to meet user needs like cost effectiveness and lower risk of injury, two important user needs for the final design. The two concepts, however, are far superior in terms of price relying on 3-D printed materials and fabrics and focus more on preventing further injury by providing ample support to the wrist and possibly finger joints. Unfortunately, due to the restrictive durability range of TPU filaments, the prototype will most likely be less durable than the reference orthotic, however being that durability isn't as important for the final design, these small flaws can be overlooked for the most part as the team focuses its design on effectiveness and cost.



Figure 5. RCAI Wrist Drop Orthosis [26].

c. Final Concept Selection

The final concept selection will not be based on a specific concept, but rather on combining the advantages of both Concept 1 and Concept 2 due to them both scoring a similar score (5-point difference). Both designs have some drawbacks. For example, the complex locking design of Concept 2 will lead to increased costs, while the installation tooling method of Concept 1 is not as efficient as that of Concept 2. This will allow the team to create the most efficient product by combining the best aspects of our previous sketches, resulting in a product that should be able to fit most the user's needs.

9. Grand Concept Design

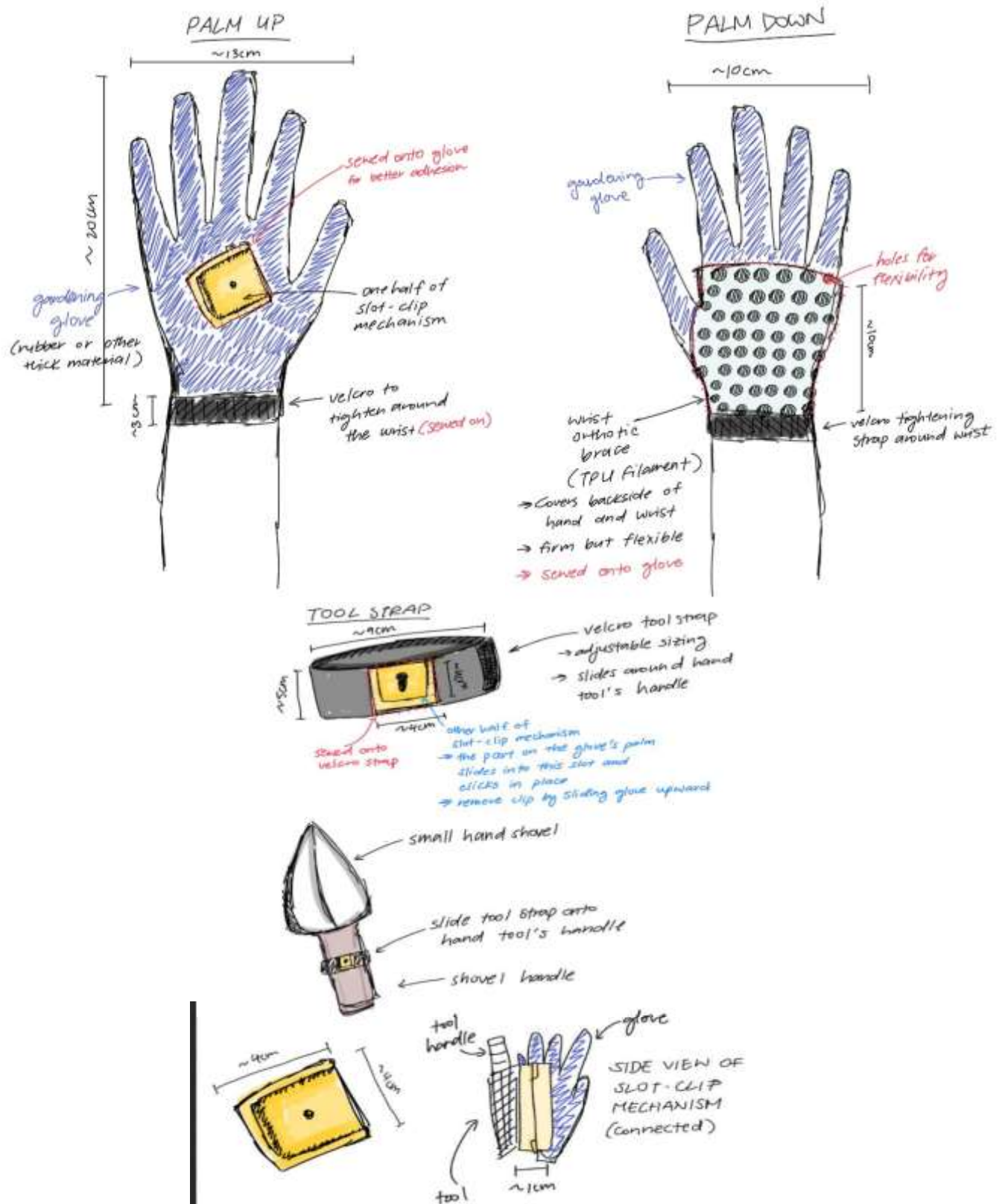


Figure 6. Grand Concept Design Sketches

The grand concept design of the team's product is a wrist orthosis that consists of a fitted brace (green brace with cutouts) for the forearm and wrist, a glove with thin metal supports running through the seams (dark blue glove), an attached clip on the palm of the glove, and an elastic band with the other half of the clip (pink band) that acts as a slotting mechanism. For demonstration purposes, the team has created a mockup cardboard shovel to show how the elastic band would be able to slide onto the handle of a small hand shovel.

The user would interact with this design by inserting their hand into the brace and glove, with the glove resting underneath the brace structure. The glove provides support to the wrist as it has strips of metal running through pockets on the inside of the glove. These supports are located on the backside of the glove. The brace will be fitted to the user using 3D modeling and will provide tight, secure support to the forearm and wrist. The elastic band will slide over the handle of a small hand shovel or tool. The clip on the elastic band will slot into the clip on the palm of the orthosis and stay in place. The user will be able to remove the clips from each other by pulling upward on the elastic-band clip, releasing it from the slot. The size of the design would be approximately big enough for an adult forearm. The team's intention with this product is to customize it personally to each user rather than utilize one-size-fits-all dimensioning. For this concept, the orthosis is modeled for a young adult male.

The features that make this design unique are the adaptability of its usage. The elastic band can be fitted onto many hand tools besides gardening tools, and the brace will provide support regardless of whether the user is using the product for a task or simply providing pain relief and compression. The anticipated challenges of developing this idea into a fully realized product are ensuring the comfortability and practicality of the design. As the team's priorities include efficiency, a challenge would be to ensure that the metal support strips do not prod the user's skin and provide support without breaking easily, remaining comfortable on the user's skin.

Additionally, another anticipated challenge is ensuring the clip can support the weight of a hand shovel. To confirm its viability and to gain traction in the market, the design must be versatile as this was the most common value that other products on the market lacked. To prove this, the elastic band must be able to fit over a variety of hand tools besides gardening tools, and the clip must work to secure the orthosis to the elastic band in each case. The product must also be able to relieve users' pain by means of support and compression across the palm, wrist, and forearm as this is the best way to test its effectiveness.

10. Prototyping Design

a. Description of Prototype

Team N plans to design and create a fully functional wrist orthotic for those who are active with hand tools and gardening instruments, reducing joint pain and preventing further injury. The brace is created based on a Surface scan (LiDAR) of a real human hand [27]. This orthotic will slip on like a glove with a rigid, weight-bearing 3D-printed

support (brace) that hugs the back of the hand; this will be sewn on to provide better durability. To wear it, the user will insert their hand into the gardening glove through the pinky-finger side of the glove; the user can close the side of the glove using a Velcro strip that runs along that side of the glove. The orthotic can then be secured using Velcro straps located at the wrist, ensuring the glove will not slip off ensuring use. In addition, a 3D-printed slot-clip is sewn into the palm of the orthosis and using a provided tool clip strap, a second slot-clip can be placed on most gardening/hand tools using an adjustable Velcro strap. When slotted into each other, the tool becomes fully fastened to the orthotic meaning the user doesn't need to strain their finger and wrist joints when gripping the tool and with the rigid support on the back of the glove, overextension of the wrist is prevented. When prototyping, budgeting and meeting our user's needs leads to the biggest limitations, scaling down previous ideas and overall simplifying the team's concept, however, with the access to 3D printers, Team N can explore more complex support designs and mechanisms allowing for design complexity at a low budget. Team N intends to use OSU's EED store to order parts such as Velcro and 3D prints and use OnShape to model testing of the orthotic.

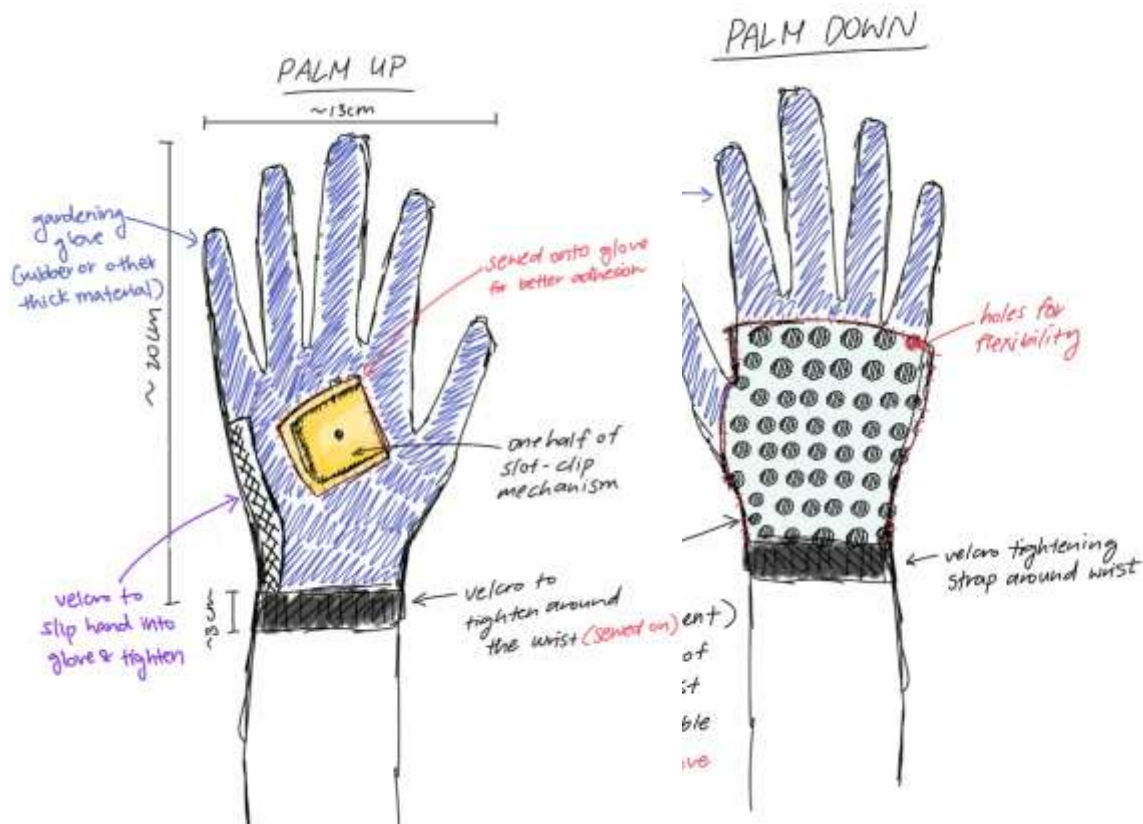


Figure 7. Wrist Orthosis Prototype Sketches



Figure 8. Slot-Clip Mechanism and Demonstration Sketches

b. Prototype Design Requirements

Design Requirement	Range of Values	Ideal Value	Design Process	Correlating User Needs
Time taken to put orthosis on	20-50 seconds	20 seconds	Test ease of putting on and taking off the orthosis during design process	Adaptiveness
Holds tools up to 10 pounds	1-10 lbs.	10 lbs.	Attach weights and test durability	Durability, Performance Efficiency
Prevents over-extension of wrist	Wrist can flex 0-90 degrees	Doesn't extend past 90 degrees	Consistently try on prototype and check wrist flexibility	Pain Reduction, Lower Risk of Injury

Fast slot-clip mechanism	10-40 seconds	10 seconds	Time how long it takes, change sizing of parts if necessary for smoother process	Performance Efficiency, Adaptiveness
Low weight	1-2 lbs	>1.5 lbs	Consistently weigh the prototype	Cost Effective, Comfortability

c. Testing Methodology and Verification Plan

To properly test and verify the design requirements for the prototype, a series of different strength, durability, and logical tests will be performed. To start, a simple load bearing test will be performed on the clip, the team wants to ensure that the quick-release system can hold up to 10 pounds of additional weight without slipping or breaking. To do this, the team will attach different weights (ex. dumbbell) to the quick-release system starting at 2.5 pounds and increasing up to 10 pounds to gradually stress testing the entire system. This process will repeat three times. Additionally, during all three testing processes, students will try to observe (ex. sound) and inspect for deformations or weak points within the system as testing occurs to ensure durability remains constant.

With the gardening wrist orthotic remains on, the team will also ask the user to bend their wrist back and forth to ensure the grading glove braces their wrist and prevents it from overextending or bending, which helps to prevent further injuries. The user will repeat this process five times to ensure its reliability and efficacy. These trials will be measured using a protractor, measuring the wrists' angle and seeing at what point it stops. Additionally, the gardening wrist orthotic will be weighed to ensure it remains lightweight, targeting under 1.5 pounds to avoid additional force on the user's joints.

Next, the orthotic will then be tested for ease of use and transfer on and off the user's hand. Using Team N members, each member will be timed for how long it takes to get the gardening wrist orthotic on and off from their hand, as well as how long it takes to attach and detach the tool onto the quick-release system's adapter. They will each be timed three separate times with their times averaged, with the team targeting a 10-second average. This is done to make sure the device is easy and adaptable to almost all users, making it a more accessible product to a broader market.

Finally, when finalizing the gardening glove, the team will create and manage a budget to ensure that the orthotic is cheap and efficient to make, utilizing processes such as 3D printing and minimizing materials for a cheaper and simpler design.

d. Correlation Matrix and Verification Scorecard

Design Requirements / User Needs Correlation Matrix								
	Time to put orthosis on	Holds tools up to 10 lbs	Prevents wrist over-extension	Speed of slot clip mechanism	Low weight	User Need Weight		
Performance Efficiency	3	3	3	9	3	1	0	1
Pain Reduction	1	1	9	1	3	4		3
Lower Risk of Injury	3	3	9	3	3	5		9
Cost Effective	0	0	0	0	9	3		
Adaptiveness	9	3	0	9	0	3		
Comfortability	3	0	3	1	9	1		
Durability	0	9	1	1	0	1		
Importance ->	52	40	88	57	66	303	<- Sum of importance values for calculation	

Figure 9. Correlation Matrix

Verification Scorecard Points Distribution Table*		
	Raw Percentage	Score Card Points (25)
Time to put orthosis on	17.2%	4
Holds tools up to 10 lbs	13.2%	5
Prevents wrist over-extension	29.0%	7
Speed of slot clip mechanism	18.8%	4
Low weight	21.8%	5
		25

Figure 10. Verification Scoreboard Points Distributed Table

Scorecard			
Requirement	Range	Score Rubric	Score
Time to put orthosis on	20-50 seconds	Lowest time = 4/4 For every 5 seconds over lowest time, -1 pts	4
Holds tools up to 10 lbs	5-10 lbs	Highest weight = 5/5 For every 1 lbs less than highest weight, -1 pt	5
Prevents wrist over-extension	Wrist can flex 0-90 degrees	Measure wrist flexibility, for every 10 degrees less than 90 that wrist can move, -0.5 pt	7
Speed of slot clip mechanism	10-40 seconds	Lowest time = 4/4 For every 10 seconds over, -1 pt	4
Low weight	1-6lbs	Lowest weight = 5/5 For every pound over lowest, -1pt	5

Figure 11. Scorecard

Item	Additional Notes	Cost
3D Printing - 2 Parts	Right Hand Support Structure, TPU	\$ 3.90
3D Printing - 1 Part	Quick Release System Holder; TPU	\$ 0.46
3D Printing - 2 parts	Quick Release System; PLA	\$ 0.86
3D Printing - 2 Parts	Support Structure and Quick Release System for Initial Prototypes	\$ 1.94
Gardening gloves	external purchase - Amazon	\$3.92
Velcro	Stock Item, 25 Inches	\$ 3.75
	TOTAL:	\$ 14.83

Figure 12. Itemized Expenses Table

e. Prototype Preliminary Design and Mock-up

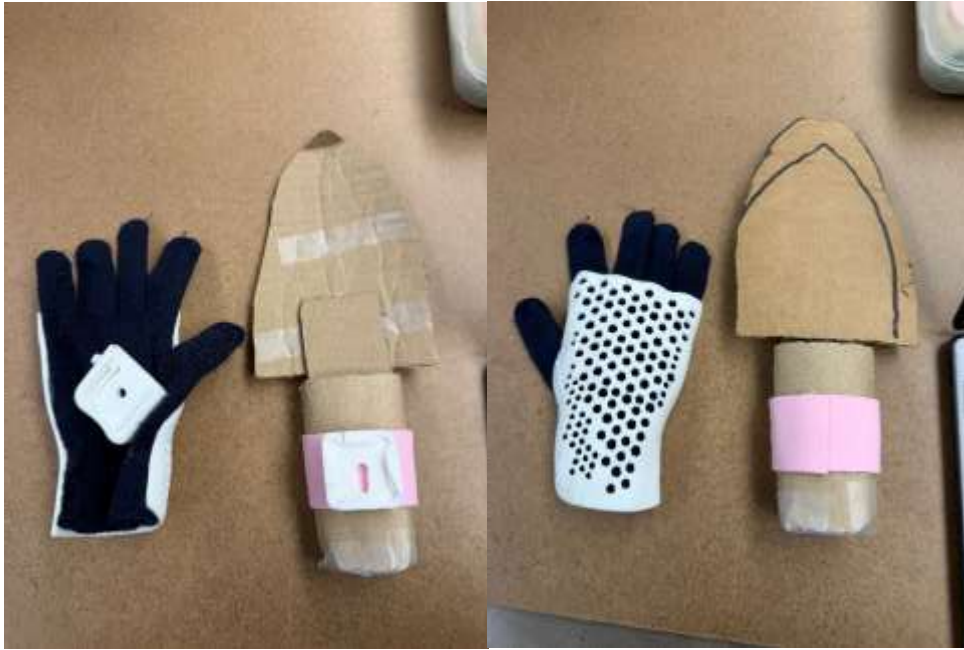


Figure 13. Wrist Orthotic and Tool Attachment Mock-ups



Figure 14. Slot Clip Mock-up Demonstration

During the creation of the mockup, the target design became clear to the team. An important change was made, switching from a magnetic clip to a 3D-printed clip system to maximize grip on the gardening tool. Utilizing a clip system ensures that the tool is securely fastened to the orthotic and it also makes the development process easier, as the clip design can be modeled in CAD software to fit the specific orthotic dimensions rather than modeling around them during the 3D modeling process and setting in magnets after. Additionally, when creating the mockup, Team N decided to exclude the original

design's finger supports, making a fingerless orthotic design as most of the joint pressure is directed on the wrist when using tools. This change makes the design cheaper and more accessible to a wider market allowing more dexterity for the user.

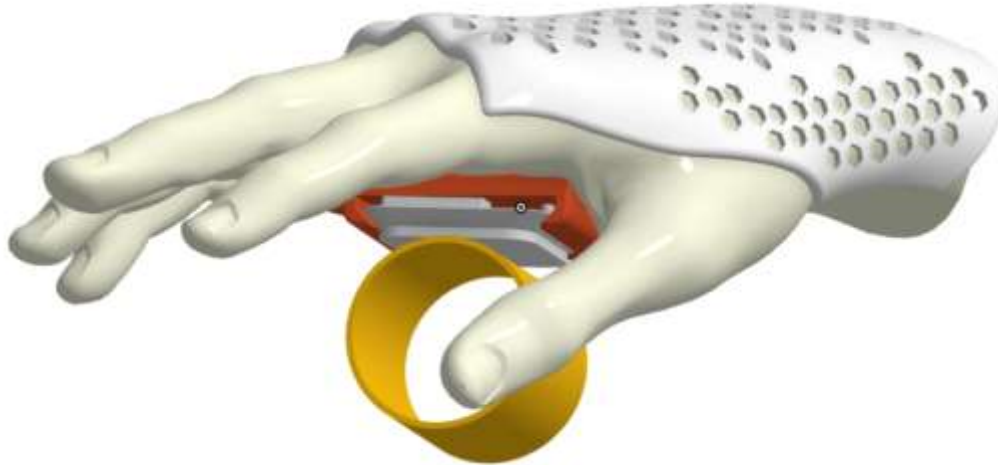


Figure 15. Sample Wrist Orthotic CAD Assembly

Detailed Design Review

11. Prototype Fabrication and Evolution

From concept to physical prototype, Team N's "Gardening Wrist Orthotic" utilized simple fabrication processes like 3D printing to create both a flexible support system for the back of the hand as well as a clip/slot system designed to secure tools to the palm of the orthotic. For the main part of the orthotic, Team N used a standard gardening glove, adhering the base clip to the palm using super glue and sewing the back-hand support to the back of the glove. Subsequently, Team N noticed that it would be nearly impossible for the user to slide their hand in the glove given the 3D-printed support hugs the wrist. To fix this, the side of the glove was cut open along the pinky side, allowing the user to fit their hand into the glove, three Velcro straps were then added on the glove, equidistant from each other, the loop side of the Velcro placed on the 3D-printed support, allowing the user to close the opening made from the cut and tighten the orthotic around their hand/wrist. Assembling the tool clip, Velcro was adhered to the 3D printed clip using super glue, formed in an adjustable loop that can be tightened to the desired circumference of the tool handle.

From here, the orthotic began its preliminary testing including a weight-bearing test, several speed tests, a wrist-extension test, and a measured test of the orthotic's overall weight. The team used OnShape to simulate a load-bearing test on the clip by applying loads up to 10 pounds on the base clip, checking for deformations or failure after each

trial. As a result, the clip system survived each trial with little to no changes, proving to be a durable clip design. Next, the team tested the speed of equipping the orthotic as well as the speed of the clip system. To do this, each member was timed putting on and tightening the orthotic, then in another test they were timed for how long it took to attach and detach a tool from the clip system. The resulting times were much less than initially expected, with the average time of these tests both being under 10 seconds, less than the expected time ranges for the trials as well. Testing the wrist overextension protection, each member put on the orthotic, bending their wrist back as far as the orthotic allowed, using a protractor to measure the wrist's angle. After every member was tested, all trials showed success in preventing the user from extending their wrist past the targeted 90-degrees, lessening the risk of further injury on the joints. Lastly, the orthotic was weighed, clocking in just under three pounds, maintaining its lightweight design while reducing additional strain on the user's hand caused by the orthotic.

After preliminary testing was complete, Team N decided that some initial design features would not need to be added to the final prototype, including metal strips that would slide between the support and the glove, providing additional rigidity to the orthotic support. Given the success of weight-bearing tests, the team concluded that the 3D-printed support would suffice, providing ample weight-distribution, reducing strain on the user's joints. On the other hand, the team added black electrical tape on the Velcro straps, cleaning up the orthotic for a sleeker design while also providing additional grip on the tool clip and the palm of the orthotic given the rubber-like properties of the tape.

12.Detailed Design

a. Final Prototype Design

Team N's final prototype design matches the team's research conducted in the beginning stages of the orthotic's development. As concluded from the research, the main joint pain caused from gardening is in the wrist joint, which is explicitly targeted by the Gardening Wrist Orthotic to resolve pain and lower the risk of further injury. Additionally, as other products on the market that relate to relieving joint pain lack adaptability, durability, and comfort, Team N strived to design the Wrist Orthotic to resolve these issues for the end users and create a product better than those present in the market. The team has designed non-standard parts to address and rectify such lacking details. The possible customization of the wrist brace makes the orthotic unique in its ability to be tailored to individuals, providing comfort. The slot clip mechanism is another non-standard part designed to improve the efficiency and adaptability of the orthotic by allowing the user to gain a firmer grip upon hand tools while gardening and having the ability to be resized and adjusted to many different handle sizes using Velcro.

Future design considerations include simplifying the brace design to allow for easier 3D modelling in the OnShape software and potentially shifting to a more versatile modelling software. Some anticipated differences between the current prototype and the final product may account for a more elegant look with improved materials rather than

the reliance upon cheap, unattractive materials – for example, using fabric or molded plastic rather than the electrical tape currently being used upon the slot clip band.



Figure 16. Final Prototype Design of the Gardening Wrist Orthotic

b. Revised Grand Concept Design

All 3D-printed components designed by Team N features weight-reducing cutout patterns (specifically, regular hexagons)—a design choice made to find a balance between the overall weight of the Gardening Wrist Orthotic and its structural integrity. The team noticed that these hexagonal cutouts significantly reduce the weight and printing costs of the 3D-printed parts (the brace and the Quick Release System) while ensuring that the components maintain a robust load-bearing structure. Furthermore, the team also learned that a gyroid internal printing structure provides maximum structural rigidity while simultaneously reducing weight. So, the Quick Release System itself was revised from its originally envisioned 100% infill density to a 50% infill density, with the internal printing structure changed from lattice to gyroid, during the 3D printing processes.

Additionally, during the prototyping phase, the team opted to eliminate the internal metal support that stays within the glove. This modification was implemented after the initial testing revealed that the glove provided support exceeding the team's expectations, even when relying solely on the TPU 3D printed brace as its structural support mechanism. Furthermore, removing the internal metal allows enhanced user comfort while wearing the Gardening Wrist Orthotic.



Figure 17. Quick Release System w/ Weight-Reducing Patterns

13.Final Prototype Verification

14.User Validation

Team N's three most important identified user needs are adaptivity, lower risk of injury, and performance efficiency, as referenced from the Pairwise Comparison Chart. The team's user validation plan to determine how well the final design of the orthotic solves the indented problem and meets the top user needs is through consistent testing and utilizing user feedback. The team will ask three individuals who fall under the target audience demographic – individuals over the age of 18 who experience temporary or chronic upper limb joint pain – to try the prototype orthotic. The team will record any given improvement advice after conducting the following tests and ask for user feedback after their experience with the orthotic to gather raw data:

1. Time how long the user takes to put on and take off the orthotic.
 - a. If the time exceeds 50 seconds, the prototype does not demonstrate performance efficiency.
 - b. If the recorded time is less than 50 seconds, the test is passed.
2. Time how long it takes for the user to put on, use, and take off the slot clip mechanism.
 - a. If the time exceeds 40 seconds, the prototype does not demonstrate performance efficiency nor adaptability.
 - b. If the recorded time is less than 40 seconds, the test is passed.
3. Measure the angles at which the user's wrist can freely bend.
 - a. If the user's wrist can bend past 90 degrees backwards, the prototype does not provide a lower risk of injury.
 - b. If the recorded degree of movement does not exceed 90 degrees, the test is passed.
4. Ask the user if they think the orthotic is comfortable or uncomfortable.

- a. If the user responds that the orthotic is uncomfortable, the prototype does not lower risk of injury, nor can it function as a competitive product on the market.
- b. If the recorded response is “comfortable”, the test is passed.

If all three users’ tests pass successfully, then the prototype will be determined as having full user validation. Otherwise, the team will continue to improve the orthotic before asking for user validation once more. The team will continue to improve, conduct trials, and ask for feedback until all three users’ tests pass.

Regarding the ethical considerations associated with collecting personal information during such trials, the team members will address each user with respect and will conduct all trials fairly, equally, and without bias. The users are not required to participate if they choose not to. Users will be given the choice to have any personal information such as their name, gender, age, and any other demographically significant detail to be recorded or not.

15.Value and Impact

Stakeholders	Economic Value	Social Value
End Users	Positive: provides a cost-effective orthotic alternative for those who want to remain active with hand tools while dealing with wrist/joint pain. Negative: N/A	Positive: opens new opportunities and activities for users, also benefits society allowing more people, especially older demographics, to perform more physical labor Negative: N/A
Investors	Positive impact: only need to invest a relatively small amount of money for a large profit Negative: recession could cause less individuals to be able to buy the product, investors lose money if product doesn't do well in market	Positive: generating intentional, measurable benefits for society, improved healthcare costs Negative: financial losses and potential reputational harm if products perform badly
Manufacturers	Positive: minimal number of parts and materials, costs less to manufacture Negative: complex design for brace customized per user makes it difficult and slow to manufacture	Positive: foster safe, diverse, and fair workplaces Negative: potential workforce exploitation, accidents in factories/workshops
Researchers	Positive: reasonable cost of orthotic, could buy many at low cost Negative: Need many test subjects to customize the brace and may take time to receive finalized products.	Positive: Demonstrate the abilities and efficiency of the orthotic to the public if research is conducted and positive review papers are published Negative: bad reviews could harm reputation of product

Figure 17. Value Matrix

Stakeholders are individuals and/or organizations with an interest or concern in the design, development, and marketing of the Gardening Wrist Orthotic. The orthotic’s most significant stakeholders are end users, investors, manufacturers, and researchers. The respective value categories that affect each of the stakeholders include economic and social impacts.

The positive economic impact upon end users is that the orthotic provides a low-cost alternative product for those who wish to remain active in using hand tools while dealing with wrist and joint pain. The team concluded that there were no negative economic impacts upon end users as the orthosis is marketed at a low cost. The positive economic impact upon investors is that such individuals or organizations would only need to invest a relatively small amount of money in the development and creation of the product in return for a large profit. However, the negative impact upon investors is that potential recession could cause a loss of the public's ability to buy the product, causing investors to lose money if the product does not perform adequately in the market. The positive economic impact upon manufacturers is that only a minimal number of parts and materials are required to create the orthotic and therefore reduces manufacturing costs, but conversely, the complex design of the 3D printed brace which is customized per user makes it difficult and slow to manufacture. Lastly, researchers could spend a reasonable amount of money to buy and test the orthotic for research purposes, but in turn researchers would require many individuals to customize the brace, causing a slow manufacturing and return process.

Team N has determined a balance between revenue and costs for the orthotic. Since the cost of manufacturing remains relatively very low, the team has decided to market the product at a price double the cost of manufacturing to ensure profits. The product will likely be self-sustaining as it was designed to be durable, and any required subsidies will be sold at a low cost to allow a large target audience to be able to purchase the product. Anticipated start-up costs will likely be expensive because 3D printers, filaments, and any required non-bulk material can be costly. Operating costs may also be rather costly as creating the CAD models for each customized brace will require a significant amount of time and experience. Forms of potential revenue will likely come directly from end users in the form of digital currency as the orthotic will only be available for purchase online. Possible subsidies required for socially focused solutions include workforce development, de-risking instruments, and program-related investments to continue funding the production and manufacturing of the orthosis.

The positive social impacts upon end users include new opportunities and activities available and allowing more individuals, especially those in the older demographic, to perform more physical labor for entertainment or professional need. There are no negative social impacts as the goals of the orthotic are to lower users' risk of injury and provide support to the wrist joint. The positive social impact upon investors are generation intentional, measurable benefits for society along with improved healthcare. The negative impact includes potential reputational harm if the product performs badly. The positive impact upon manufacturers includes the ability to foster safe, diverse, and fair workplaces, while the negative impacts upon this stakeholder are potential workplace exploitation and the risk of accidents at manufacturing sites. Finally, the positive social impact upon researchers is their ability to demonstrate the orthotic's efficiency to the public and publish positive review papers in its favor, boosting the orthotic's sales while reaching a larger audience. The negative social impact upon researchers is the possibility of bad reviews that could harm the product's reputation and/or possible bias when conducting research upon the wrist orthotic.

Some potential unintended negative consequences of the team's solution on society or the environment include excessive filament waste when 3D printing the brace and slot clips, as support structures may not always be recyclable and often, recyclable filament itself can be costly. The required sanding, frequently required testing, and potential failed prints may rack up production costs. Additionally, the brace may not provide the exact support every individual user's needs as such joint pains can be arbitrary and inconsistently afflicted upon the user, causing the brace to be a general solution rather than an extremely targeted one. Since the team is utilizing scanning software to shape the brace for each user, the scans may not be perfectly accurate for the user and may require multiple trials, increasing the time the end user must wait for their order to be completed.

16. Project Recommendations and Next Steps

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Appendix A. Evidence of Brainstorming

1. Discovering intended use/users.

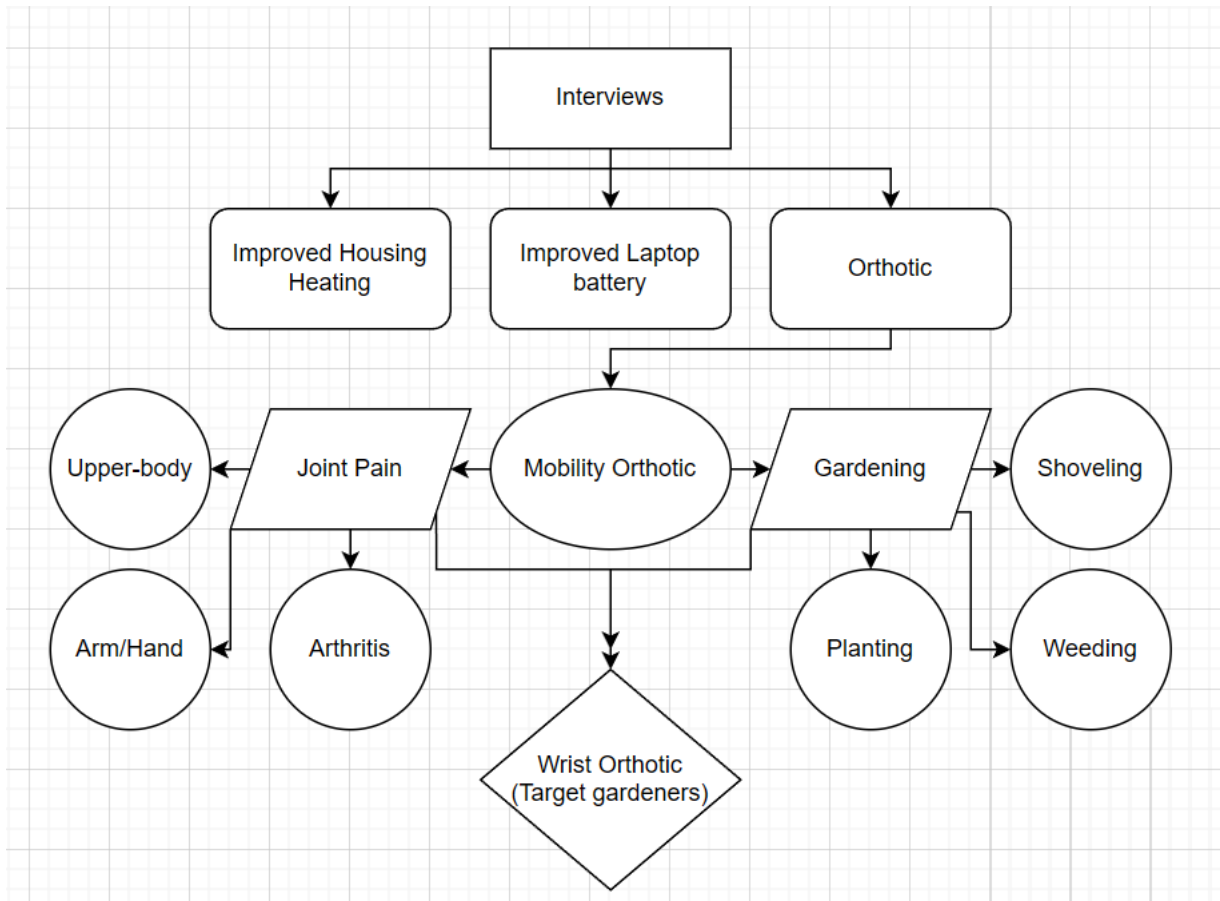


Figure 6. Discovering Intended User/Users Flow Chart

2. Brainstorming Notes:

- Research half about how to build/design, half about joint pain/mobility
 - Each person finds 3 sources for the table
 - Magnets / Quick-release systems to switch out accessories, Velcro, adjustable rods
- Overall product idea: gardening mobility aid for arm with force boost mechanism

3. Design Feedback:

- Cost for a customized wristband
- Creating an elastic band with a magnet inside that can attach/slip onto tools --> can connect to a magnet on the glove
- Easier to use Velcro than twist knobs (?)
- Swivel on tool attachment

4. Device Concepts:

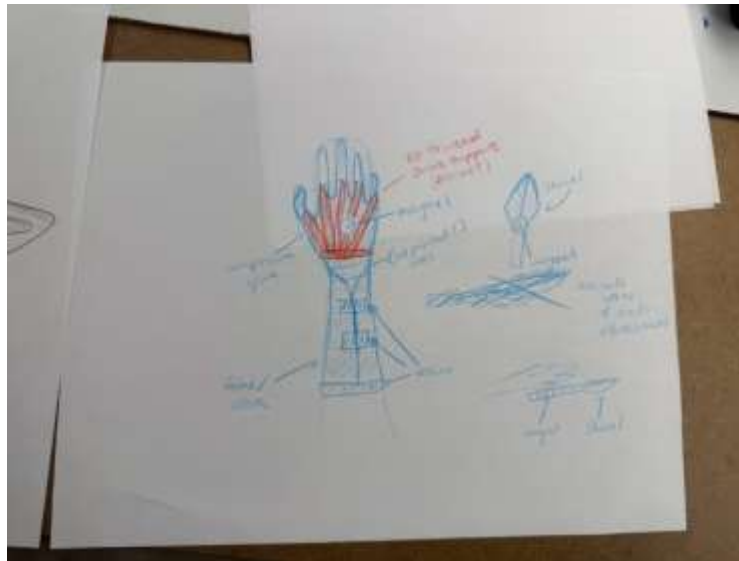


Figure 7. Concept 1 Sketch

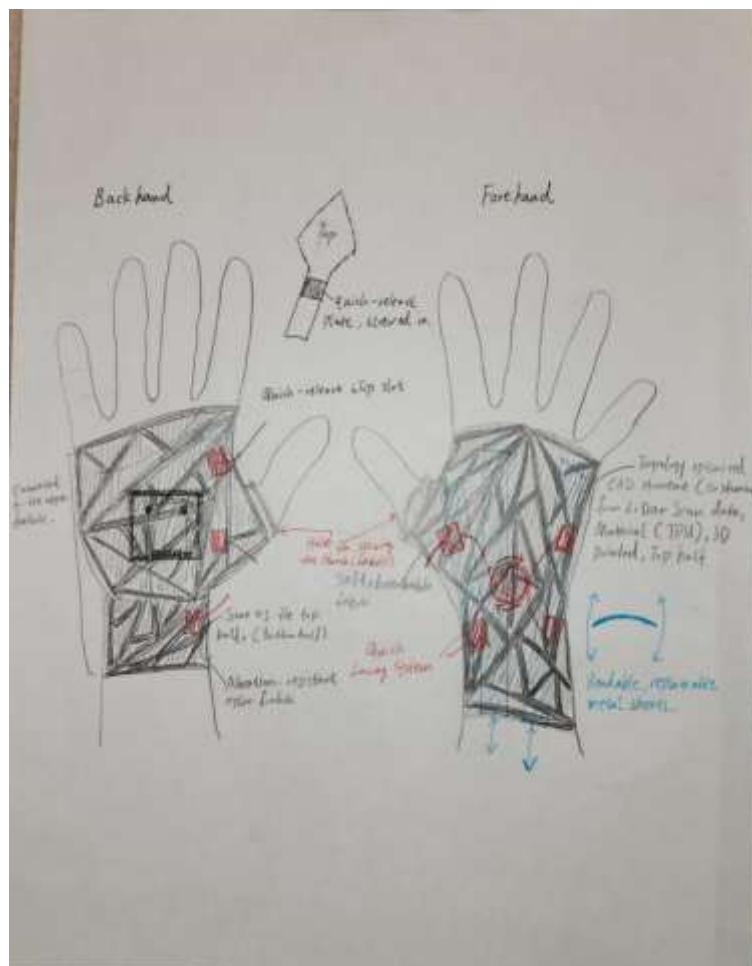


Figure 8. Concept 2 Sketch



Figure 9. Concept 3 Sketch

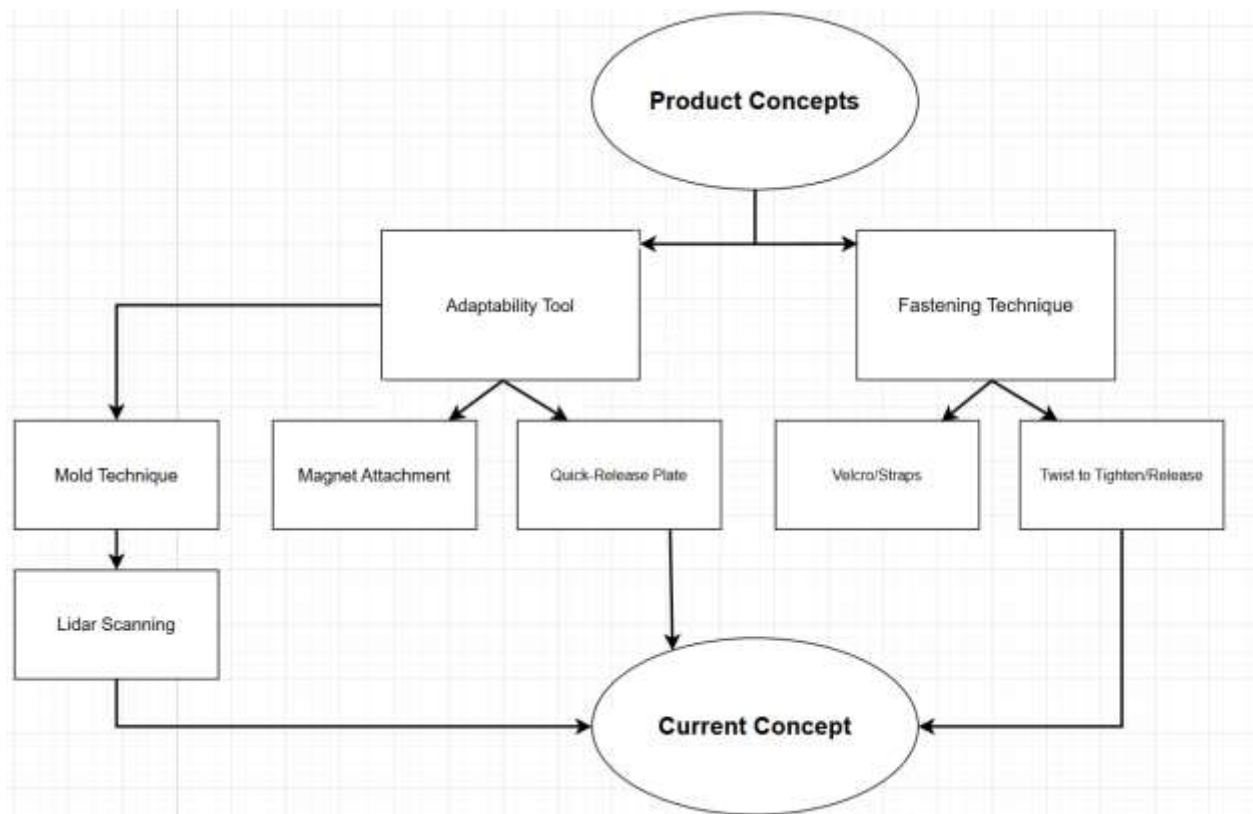


Figure 10. Team's Product Concept/Idea

1. Grand Concept Design

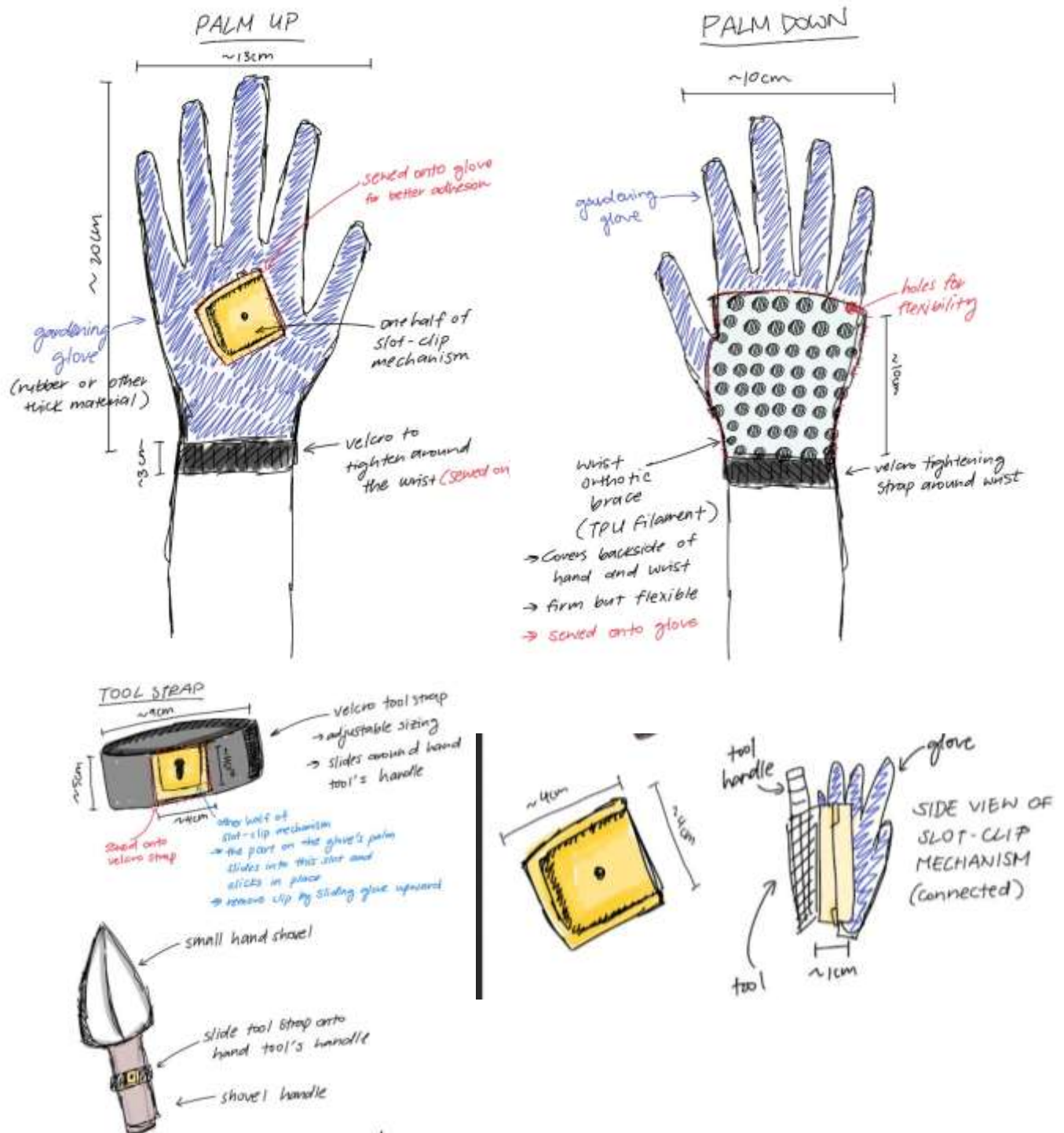


Figure 11. Updated Sketches of Grand Concept Design

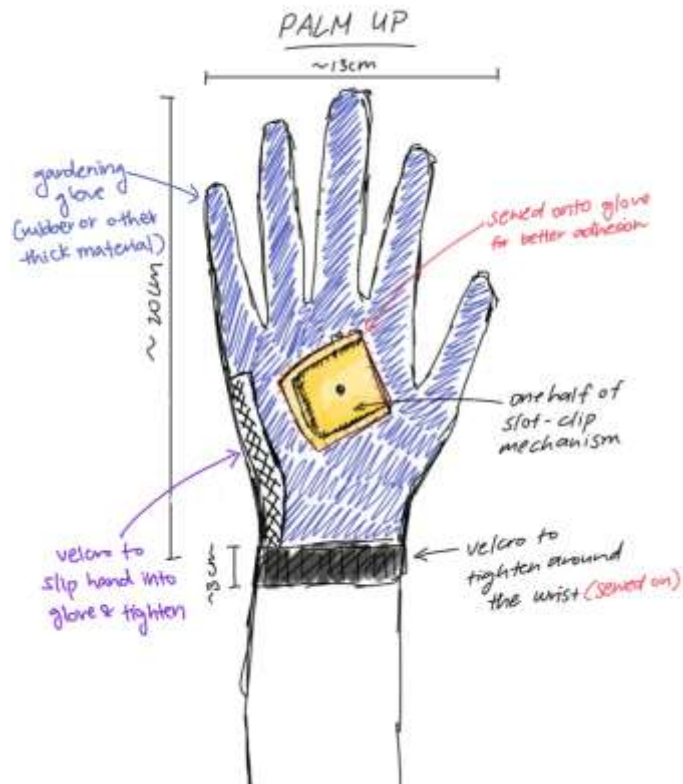


Figure 12. Updated Sketch of Wrist Orthosis with Velcro on Side of Glove

Appendix B. Research Methodologies

Interview 1:

1. *Can you tell me a little bit about yourself (age, ethnicity, gender, education, socio-economic status, ability, household make-up, etc.)?*
 - a. Indian American female, age 50, mother of two, “empty nester”, college education, married
2. *What do you do for a living?*
 - a. Stay at home mother for the past 18 years
3. *What are some of your hobbies?*
 - a. Gardening, sewing, DIY projects, interior decorating
4. *What does your typical day look like? What are some tasks you perform throughout your daily routine?*
 - a. Cook for family, clean/organize the house, walk the dog, garden in the front and backyard
5. *What are some products you use regularly in your daily tasks and hobbies?*
 - a. Paper towels, shovels, watering can, pots/pans, spatulas, laptop, phone
6. *Can you describe some things that bother or frustrate you when performing your routine tasks?*
 - a. Chronic joint pain due to tennis elbow and arthritis, consistent pain in wrists
7. *Is there anything that would make it easier or more enjoyable to perform these tasks?*
 - a. A kind of brace or product that could put less strain on arms when gardening
8. *Of the problems/frustrations/pains you’ve described, are there any that really stand out?*
 - a. *Why is this a significant problem?*
 - i. Weeding is difficult because plants must be reached from the roots, but this puts pressure on the joints and causes more pain
 - b. *What are you currently doing to alleviate this problem?*
 - i. Taking pain medication, using home remedies/ointments
 - c. *What do you like about how you currently handle this problem?*
 - i. Easier to work after ingesting medication/resting
 - d. *What isn’t working about how you currently handle this problem?*

- i. Can't be efficient about the weeding, needing to take excessive breaks, strenuous on the arms and lots of pain afterward
- e. *What products do you wish existed that currently do not?*
 - i. Some sort of arm brace that can keep arm steady and/or a tool that can reach the roots of small weeds without needing to overuse wrists/elbow

Summary:

The interviewee is a female of age 50 who is a stay-at-home mother of two children that currently attend college. She is married, has a college level education, and has previous work experience. In the interview, she said her hobbies include gardening, sewing, and craft DIY projects. We discussed what her typical day looks like and whether anything bothers her when working on her hobbies or daily tasks, to which she stated that her chronic joint pain stemming from arthritis and tennis elbow cause severe pain along her arm when she gardens outside. When asked about specific products she would find useful to her problem, she responded that a brace or structural support item would help her with weeding, which requires her to reach the root of the plant and is one of the more difficult aspects of gardening and a large contributor to her pain. Additionally, she mentioned that a tool that could reach the weeds' roots without requiring her reach as far would be beneficial. The interview went well, as I received good information about a potential problem many people would have, and therefore a starting point for product brainstorming.

The user is a woman in her 50s who has tennis elbow and arthritis.

Task: Weeding the vegetable garden

Pains:

- It can be extremely painful to pull a large number of weeds.
- It is difficult to reach the roots of the weeds buried in the soil.
- It is often necessary to take many breaks and take pain relief medication excessively.

Task: Managing clutter and organization

Pains:

- It is costly to keep buying organizing shelves
- There is a lot of clutter to organize on a daily basis
- It is difficult to keep things in the same place and remember where the items are

Figure 13. Interview 1 Analysis

Interview 2:

Getting to know your user (~5 mins)

1. *Can you tell me a little bit about yourself (age, ethnicity, gender, education, socio-economic status, ability, household make-up, etc.)?*

- I am Enid, a sixty-year old woman who works in publishing, that now works remotely from home.

2. *What do you do for a living?*

- I am an executive in publishing children's books.

3. *What are some of your hobbies?*

- I like to garden, run, cook, birdwatching, and going to antique stores.

4. *What does your typical day look like? What are some tasks you perform throughout your daily*

routine?

- I wake up around 5:30 in the morning, clean the kitchen and make coffee for myself, I plan my day while I drink my coffee. I then stoke the fireplace to heat the house and check into work on my computer, and I try to clean at least one room a day, generally trying to keep the house clean throughout the week. Since I recently moved to an older home, I am currently in the process of renovating and decorating it in recency as well.

5. *What are some products you use regularly in your daily tasks and hobbies?*

- I use all-purpose cleaners for the house, I use wood oil for the floors, I use a Shark vacuum and robot vacuum for the carpets, I use my laptop/iPad for work as well as an Eero router for Wi-Fi., and I also use a coffee maker for my coffee.

User Experiences (~10 mins)

6. *Can you describe some things that bother or frustrate you when performing your routine tasks?*

- I get frustrated with Wi-Fi issues in general as we live in a more rural community, I also run into issues with our robot vacuum as it gets stuck frequently and doesn't properly map the house to reach optimal cleaning locations. I also have problems with our hot water is it takes at least 5 minutes to heat up, making cleaning and cooking take longer. More recently I have had trouble with our coffee maker as it doesn't properly grind the beans fine enough most of the time.

7. *Is there anything that would make it easier or more enjoyable to perform these tasks?*

- What would make heating the house and improving insulation easier would be to get a new heating system for the house so that a fire doesn't need to be stoked every so often. A solution for my coffee maker would be getting a new coffee maker or a separate coffee grinder to ensure that the coffee powder is fine enough for my daily brew.

8. *Do any of the products you use lack certain features or could those products be improved?*

- What would be improved is our heating system, our Wi-Fi, my robot vacuum and the coffeemaker.

9. *Thinking about your hobbies, is there anything that you find difficult or don't enjoy?*

- I do not enjoy cleaning out our pond, I also do not enjoy clearing our land from ivy and brush as it grows too much and becomes difficult to trim especially in the winter with all the snow.

10. *Of the problems/frustrations/pains you've described, are there any that really stand out?*

- Heating

a. *Why is this a significant problem?*

- The issues with the heating affect a bunch of things around the house, causing issues with our piping and making morning showers and the use of hot water in general just take longer.

b. *What are you currently doing to alleviate this problem?*

- I am planning on getting a professional to evaluate what must be done to improve the heating, possibly having to buy baseboard heaters and replace our fireplace.

c. *What do you like about how you currently handle this problem?*

- I enjoy the comfort of the fireplace that we currently use and it makes the room that it's located in a much more relaxing area.

d. *What isn't working about how you currently handle this problem?*

- The fireplace doesn't sufficiently heat the entire house enough, only really heating the first floor.

e. *What products do you wish existed that currently do not?*

- I wish that there was a historically sensitive heating system that would be both efficient and aesthetically pleasing in a much older house with exposed beams and that sort.

Summary

To identify common issues and understand the “users” in an engineering task, Enid was interviewed, she is a 60-year-old mother who works as an executive in publishing for children's books. Enid is an active person who loves to garden, run, cook, and birdwatch in

her free time. She recently moved to an older rural farm which is the biggest frustration for

her with furnishing and fixing it up. Revealed in the interview is that daily, Enid wakes up

early, makes her coffee and cleans the house, aiming to clean at least one room a day to maintain a tidy living space. She also stokes the fire every morning before checking into work on her computer and throughout the day to heat the house. The age of the house as well as outdated appliances make her routine a bit more difficult in the morning she told, as internet issues, problems with heating, as well as issues with the robotic vacuum and the coffeemaker. Her biggest issue being that of the heating, the fireplace not providing sufficient heat for the entire house, leaving some areas colder than others. Overall, the interview was organized, well-paced and in the future recording the interview, with consent

allows me to review the interview and recite exactly what was said by the interviewee.

Task: Heating the house

Pains:

- The fireplace doesn't heat the entire house, making some rooms cold
- Placing the log in the hot fireplace can be dangerous and lead to injuries or damage to the house
- Wood needs to be bought and split beforehand, making more tedious prep work

Task: Cleaning the house

Pains:

- The hot water takes a long time to heat up, making cleaning longer
- The age of the house can make it dangerous and difficult to clean certain spots in the house
- The robot vacuum frequently gets stuck and incorrectly maps the house, not working to its best potential.

Figure 14. Interview 2 Analysis

Interview 3:

1. *Can you tell me a little bit about yourself (age, ethnicity, gender, education, socio-economic status, ability, household make-up, etc.)?*

I'm a 19-year-old first-year undergraduate student (Majoring in Computer Science) from China (an international student) and an only child.

2. *What do you do for a living?*

I'm a student who lives off my parents' money.

3. *What are some of your hobbies?*

Playing badminton, working out, playing video games

4. *What does your typical day look like? What are some tasks you perform throughout your daily routine?*

After class, I do my homework and then play games. In the evening, I go to bed after working out. On weekends or holidays, I might play games all day or sleep all day.

5. *What are some products you use regularly in your daily tasks and hobbies?*

Using a computer (to play games) and using a badminton racket (to play badminton).

6. *Can you describe some things that bother or frustrate you when performing your routine tasks?*

Playing games or studying outside the dorm often results in insufficient battery life for my computer, which runs out of power easily. However, carrying a charger is too heavy.

7. *Is there anything that would make it easier or more enjoyable to perform these tasks?*

Bring a power bank; this will power your laptop and reduce the weight of carrying the charger.

8. *Do any of the products you use lack certain features or could those products be improved?*

Use batteries with higher energy density and chips with higher performance and lower power consumption for computers.

9. *Thinking about your hobbies, is there anything that you find difficult or don't enjoy?*

My computer keeps crashing while I'm playing games, and the badminton rackets I rented from school are of very poor quality and very heavy.

10. *Is there anything else that you find difficult or are unable to do, but wish you could?*

My computer crashed, but I don't know how to fix it or troubleshoot the problem (computer repair).

11. *Of the problems/frustrations/pains you've described, are there any that really stand out? (Choose at least one)*
a. *Why is this a significant problem?*

When a computer crashes, users don't know how to handle it, leaving them without a computer to play games or study, thus delaying their learning/game progress.

b. What are you currently doing to alleviate this problem?

Learning some computer knowledge can alleviate or resolve these problems to some extent, thus preventing them from significantly impacting the user's life.

c. What do you like about how you currently handle this problem?

Users can fix their computers without relying on others, while also saving potential computer repair expenses.

d. What isn't working about how you currently handle this problem?

I haven't learned enough and have limited resources, and I still can't handle more complex computer problems on my own.

e. What products do you wish existed that currently do not?

Software that can automatically troubleshoot and resolve computer system/hardware problems, and a computer with long battery life and reliable performance.

Summary:

The user interviewed today is a 19-year-old male Chinese student currently a freshman majoring in Computer Science at Ohio State University. He is an only child and is currently fully supported by his parents. As a student living on campus, his daily life mainly consists of attending classes, doing homework, and then playing games or going to the gym/playing badminton after finishing his homework. On weekends or holidays, he might play games all day or sleep all day. He frequently uses his laptop for studying or playing games.

The main purpose of this interview was to understand Kevin's daily life and the difficulties he encounters with some of the products he frequently uses, such as his laptop and sports equipment. The interview went smoothly because the interviewer provided detailed and thoughtful answers, particularly regarding his laptop's performance, malfunctions, and battery life. The focus of the interview was on the interviewee's dependence on his computer, especially during study and gaming. The user also emphasized problems with his computer system (black screen/freezing) when gaming or studying. If a follow-up interview were conducted, I would need to further explore the emotional impact of these issues, and understand the specific frequency of these problems to better understand the user's pain points, their severity, and persistence.

Task 1: Using a laptop for studying and gaming outside the dorm

- Laptop battery life is insufficient, especially during extended use.
- Carrying the power adapter adds significant weight and is inconvenient.
- Laptop performance is unstable, and the system frequently crashes.

- When the laptop malfunctions, both studying and entertainment activities are inevitably delayed or interrupted.

Task 2: Leisure activities (playing games and badminton)

- Laptop freezing while gaming degrades the user's gaming experience and affects game progress.
- Users lack the knowledge to independently diagnose and repair hardware and system problems.
- The badminton shuttlecocks provided by the school are heavy and of poor quality.

Interview

4

Questions Asked and Answers:

Can you share some information about yourself such as your age, gender, and Education? I am a 53-year-old male with a high school education.

What do you do for work? I am a custodian at a high school.

What are some hobbies you enjoy? I enjoy fishing, hunting, and racing.

What does your typical day look like, and what are some tasks you perform in your daily routine? My day mostly consists of cleaning in which I sweep, vacuum, and mop
37
classrooms.

What are some objects or products that you may find yourself using in your daily life

Whether it's a task or a hobby? Backpack Sweeper, Chemicals, Fishing Poles, Guns.

User Experiences:

Can you describe any things that tend to bother or cause you to struggle when performing your daily tasks? When supplies aren't stocked or where they are supposed to be, when equipment doesn't work as it should, and faulty equipment.

Is there anything that could make those tasks easier or more enjoyable to perform?
When equipment works properly without any hassle.

Do any products that you use to complete your tasks lack features or could be improved? Making things weighs less and making equipment easier to carry.

About your hobbies, is there anything you find difficult or unable to complete but wish

you could do so? Knee pain causes me issues; I wish I could climb or venture out further.

Is there anything you find a struggle or are completely unable to do in which you wish

you could?

I struggle with mobility; it would be nice to get around easier.

Of the struggles/frustrations that you've described in the previous questions, are there really any that stands out? Why is this a significant problem and what products

do you wish existed that currently do not? Primarily just equipment being lighter, but the

school doesn't want to pay for better equipment.

Description/Summary/List:

The interviewee was a 53-year-old man who works as a high school custodian who spends

most of his time either cleaning classrooms, or interacting with his hobbies which are fishing, hunting, and racing. The questions revolved around what he does and what problems he encounters when he's doing his tasks. Most of the problems he mentioned were about his lack of mobility as well as equipment that could be easier to carry in different ways. Overall, the team believes that the interview was a success but, in the future, the team could word the questions in a better format so it's more interactive rather than just answering this and answering that. All around the team could just interact with them more and make it a more enjoyable experience for them.

Task: Carrying around equipment to numerous classrooms to clean

Pains:

- The equipment can be heavy, which causes physical stress and exhaustion.
- The equipment can be difficult to carry, which can relate to other things other than just the weight such as handles and if it's awkward to carry.
- Equipment can be missing, not working, or faulty.

Task: Doing outside hobbies such as hunting and fishing.

Pains:

- Movement restrictions due to previous injuries and joint pains.

- Arms and wrists are not working correctly due to overstressed joints, muscles, and bones.
- Actions cannot be done as he wants because he is unable to do them himself due to back pain and difficulty bending over.

Appendix C. Prototype Working Drawing Packet

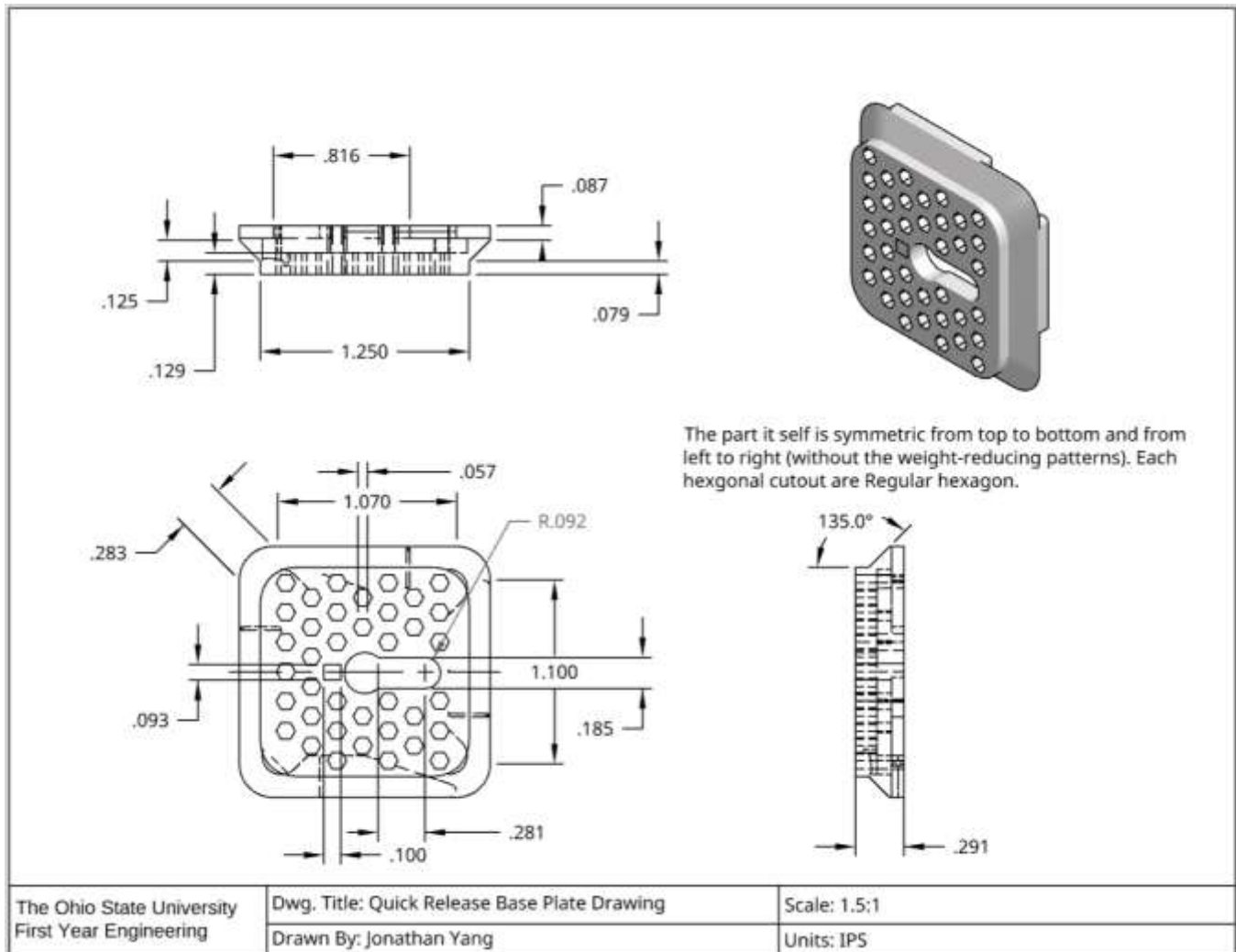


Figure 14. Quick Release Base Plate Drawing [28]

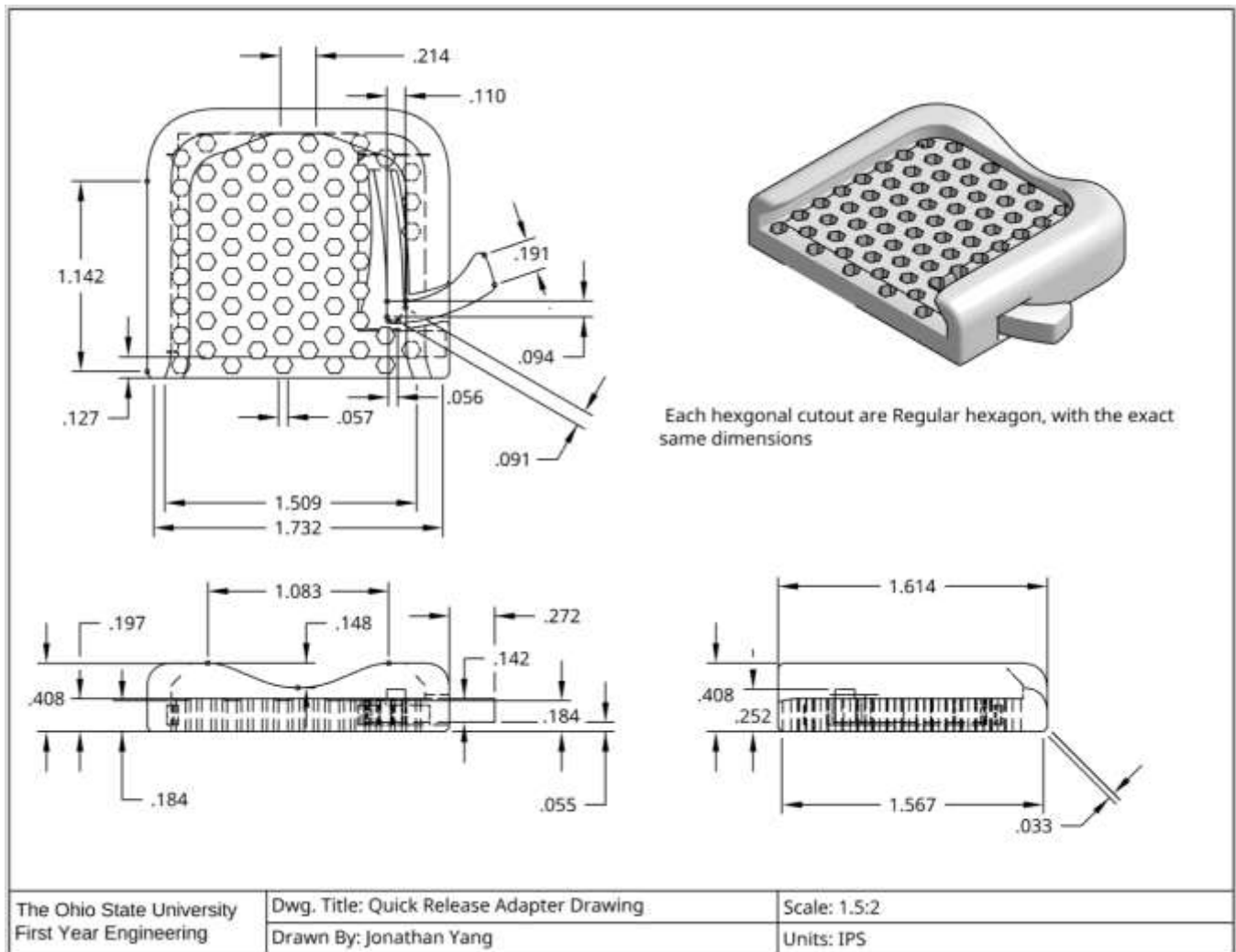


Figure 15. Quick Release Base Adapter Drawing [29]

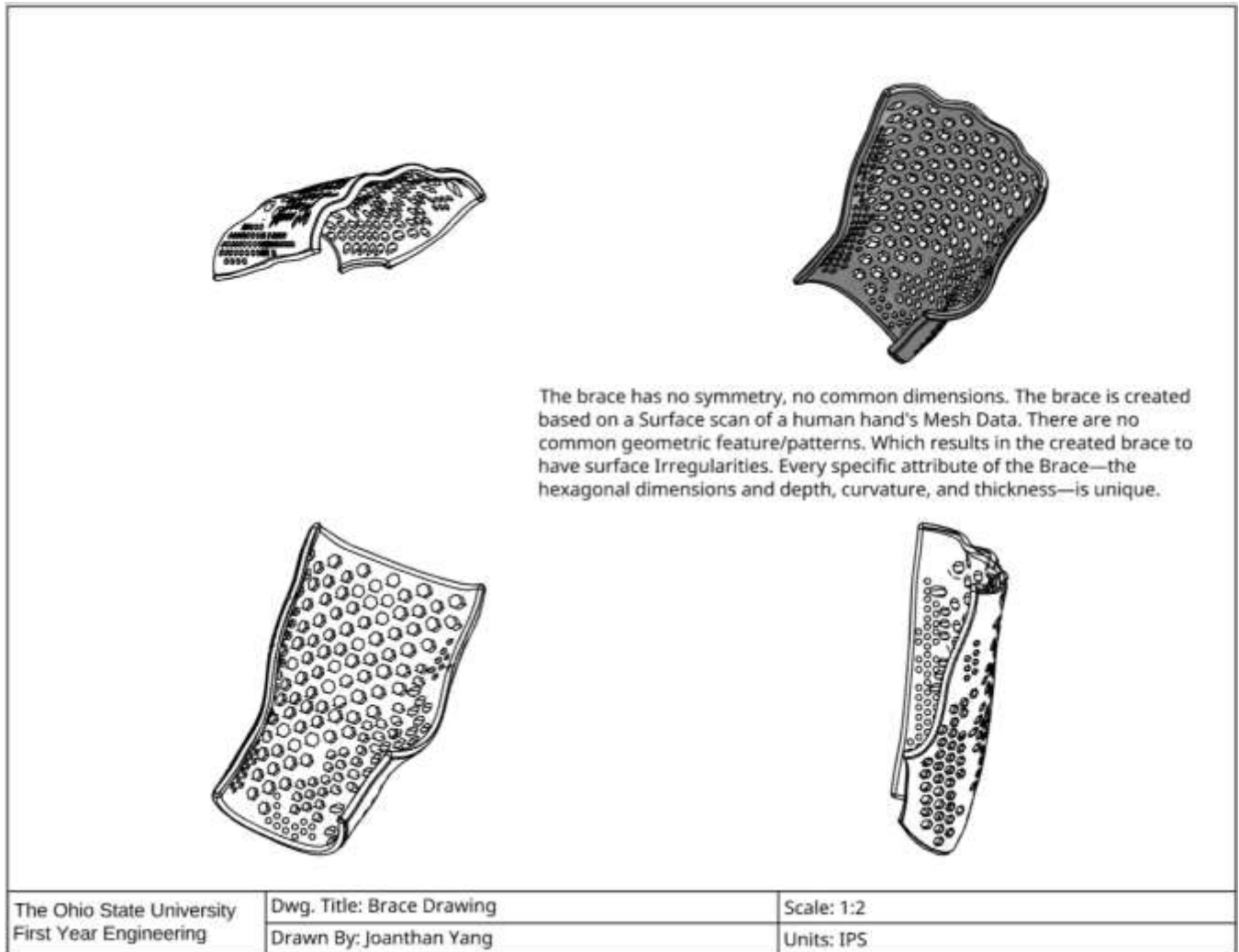


Figure 16. Quick Release Base Adapter Drawing

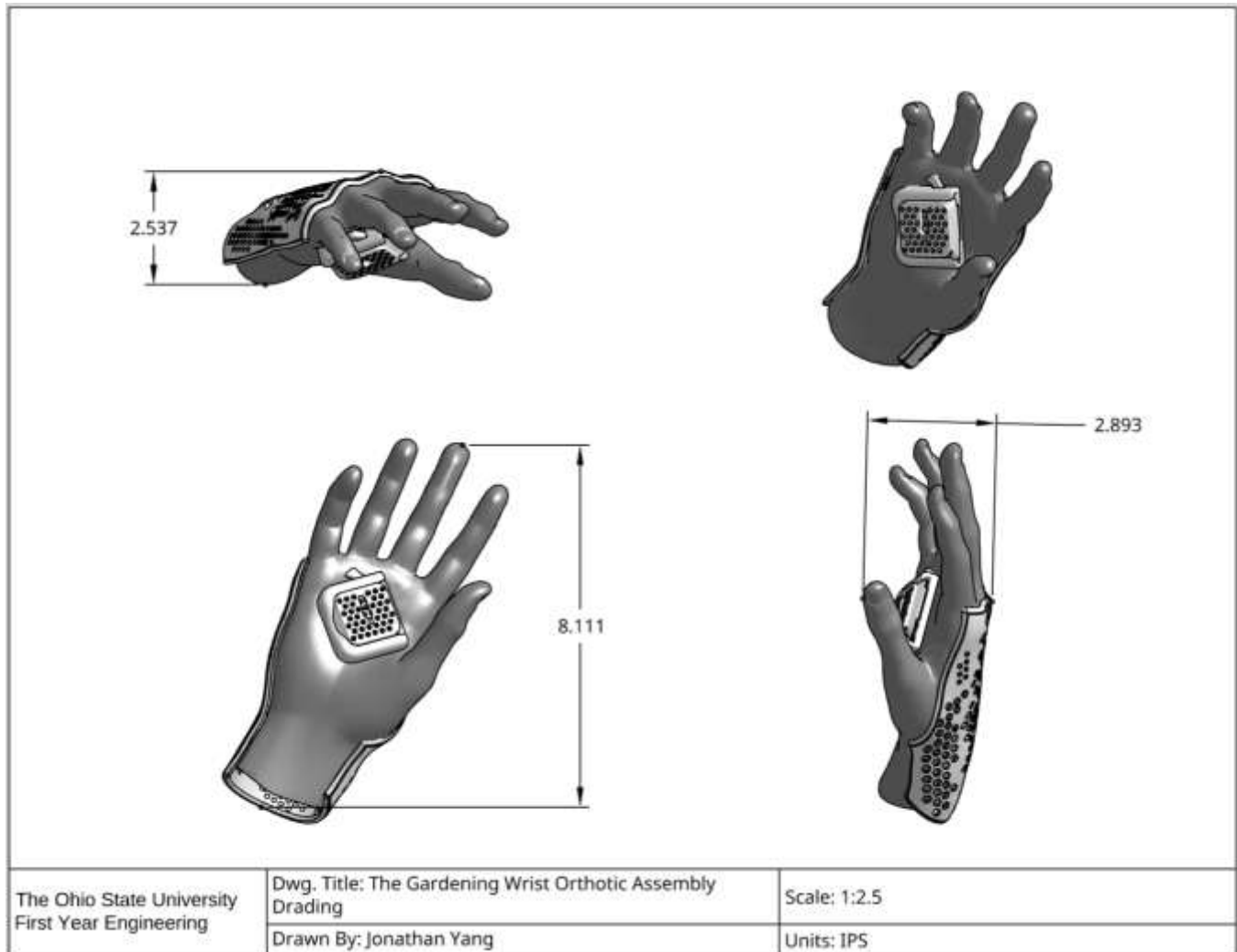


Figure 17. The Gardening Wrist Orthotic Assembly Drawing

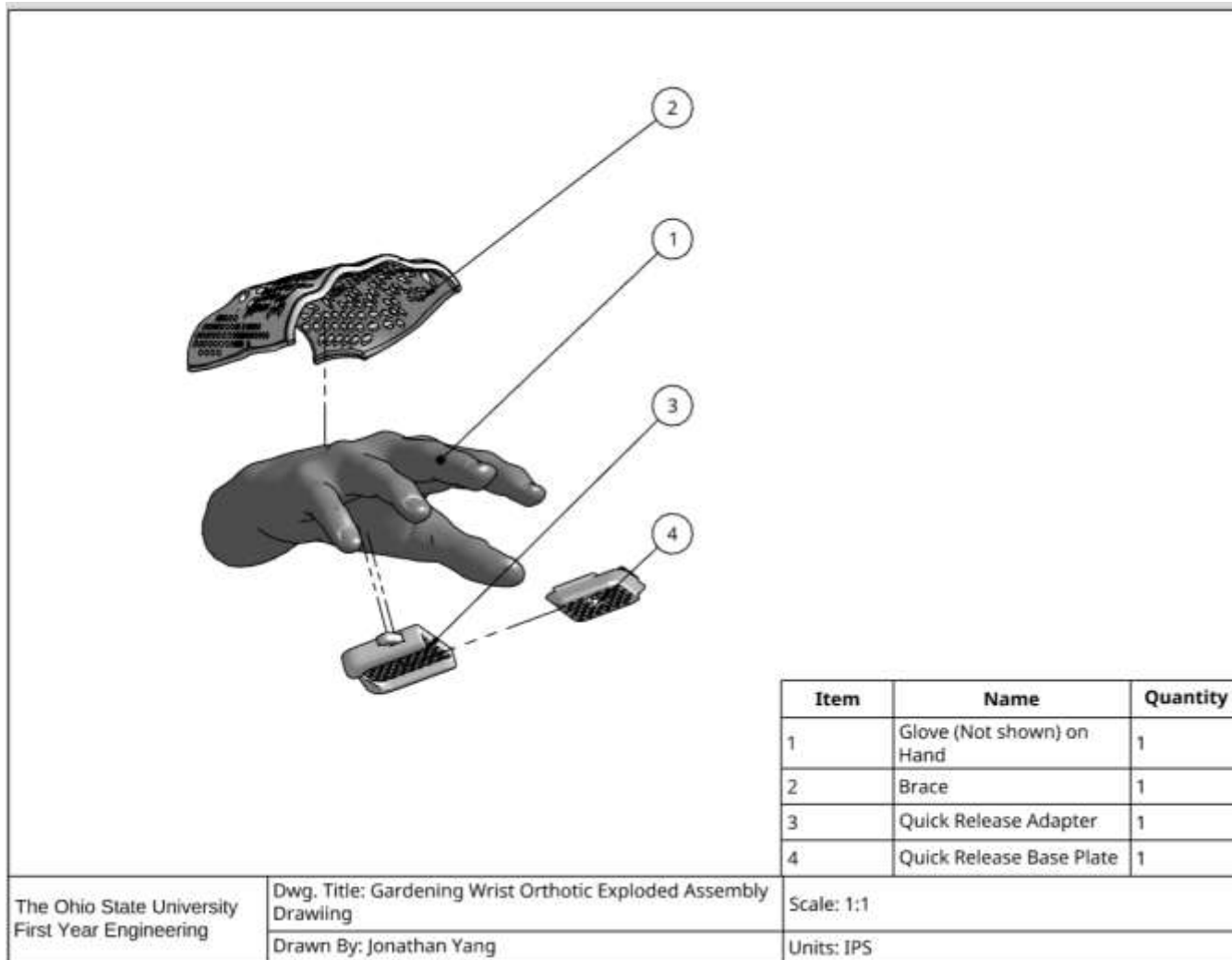


Figure 18. The Gardening Wrist Orthotic Exploded Assembly Drawing

Appendix D. Additional Raw Data

OnShape Simulation Data:

Testing Day 1 Data:

Test	Data	Result
Slot clip speed	4.3 seconds	Passed
Putting on orthosis speed	8.65 seconds	Passed
Load bearing test	25 lbs	Passed
Wrist extension test	-90 to 90 degrees	Passed
Weight test	0.55 lbs	Passed

Appendix E. Project Management Schedule and Meeting Minutes

Team N's main goals and deliverables for this design project are given as a process; by specific deadlines, the team expects to fully CAD, print, assemble, and test the wrist gardening orthotic, ensuring that it meets all user needs and determined design requirements. Overall, the team's mission is to complete a fully functional orthotic prototype that delivers on all intended applications.

2. Project Schedule

a. PERT Schedule

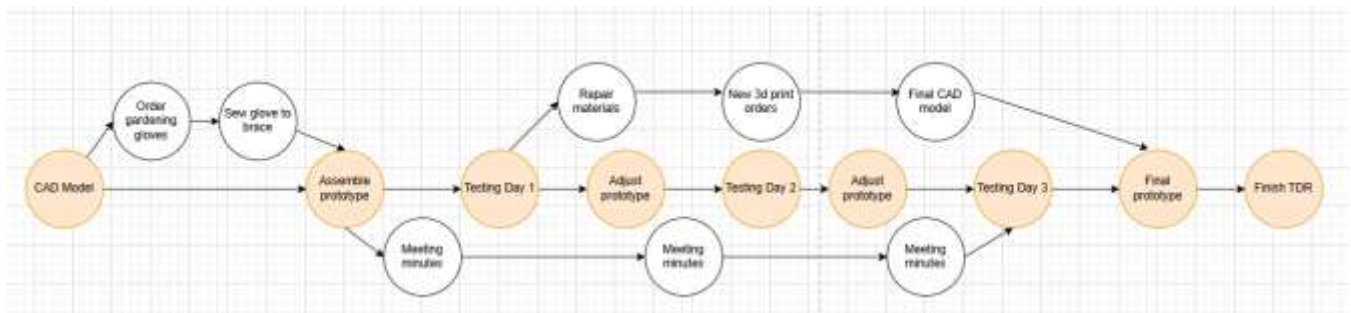


Figure 19. PERT Chart: Critical Path of Deliverables and Tasks

b. Gantt Chart

Detailed Design Phase

Phase start date: Mon, 4/6/2026

SIMPLE GANTT CHART by Vertex42.com
https://www.vertex42.com/ExcelTemplates/simple-gantt-chart.html

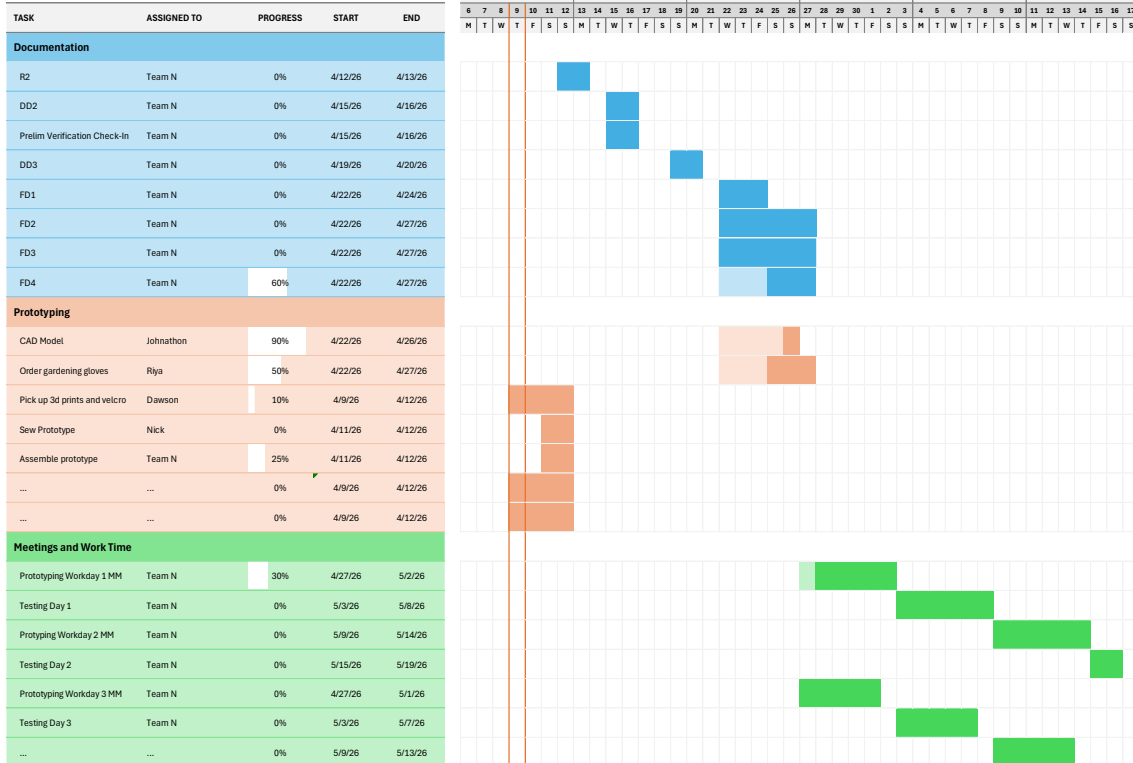


Figure 20. Gantt Chart: Schedule of Final Deliverables and Tasks

3. Prototyping Workday 1 Meeting Minutes

Meeting Minutes 1 – April 10, 2026, 3:55, 1182 Lab Hitchcock Hall 308, Riya, Jonathan, Nick

Objective statement: Purpose of the meeting was to begin assembling collected parts of the orthosis and update meeting.

Completed tasks:

- Original Gantt Chart and PERT chart finished
- Last 3d printing and EED store orders put in
- Materials collected
- Next meeting time established
- Riya: assemble the slot clip mechanism and Velcro strap for tool handle

- Nick: sew brace onto gardening glove, add Velcro to wrist and side of the orthosis
- Jonathan: work on CAD model and collect 3d prints and orders
- Status: slot clip mechanism finished, brace 50% sewed onto glove, wrist tightening strap added

Tasks to be completed:

- Jonathan: Work on CAD model, put in new orders if needed
- Riya: Complete meeting minutes 1
- Team N: Update and work on the TDR
- Finalize scorecard & complete CATME evaluations

Project timeline (updated Gantt Chart):



Figure 21. Updated Gantt Chart: Schedule of Final Deliverables and Tasks

Decisions:

- Team N will meet on April 11, 5pm at Thompson Library to continue assembling the prototype
- Finish CATME evaluations
- If needed, meet on Sunday to finish iteration 1 of prototype

4. Prototyping Workday 2 Meeting Minutes

**Meeting Minutes 2 – April 20, 2026, 4:10 PM, 1182 Lab Hitchcock Hall 224,
Jonathan**

How did feedback from users influence your plan?

Users noted that securing tools via magnetic attachment could be constrained by the tools' material and the strength of the magnetic; consequently, the team opted for a quick-release system design, which features a lower barrier to use and lower production costs.

Objective statement: Refine the appearance of the Gardening Wrist Orthotic, complete CAD Drawings/Simulations

Completed tasks:

- CAD Drawings/Simulations
- Refined the Gardening Wrist Orthotic
- Jonathan: Refine the appearance of the Gardening Wrist Orthotic, complete CAD Drawings/Simulations
- Status: Gardening Wrist Orthotic is fully completed (could expect possible minor adjustments in the future).

Tasks to be completed:

- Team N: FD1, FD2, Design Showcase Flyer
- Complete Final CATME evaluations

Project timeline (updated Gantt Chart):

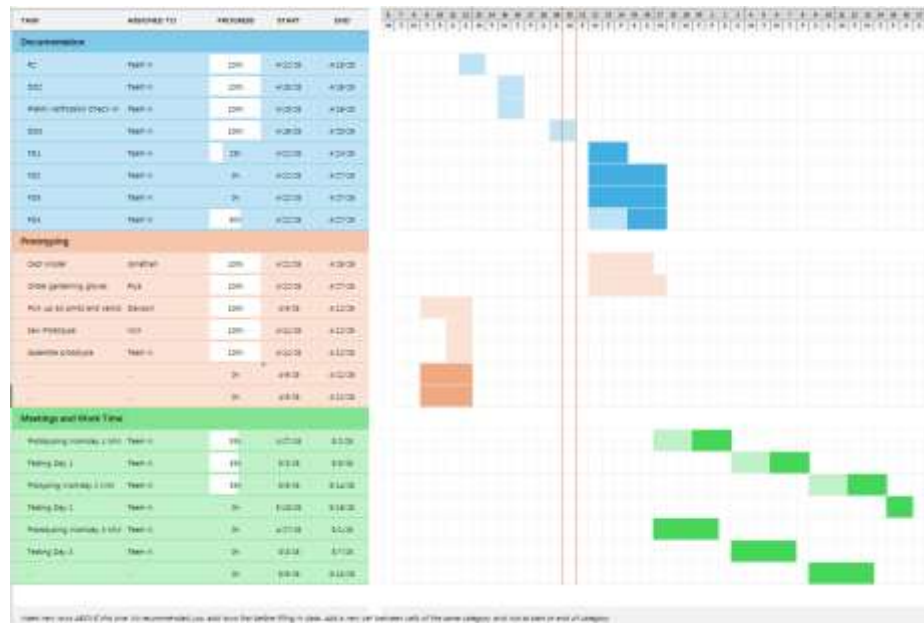


Figure 22. Updated Gantt Chart: Schedule of Final Deliverables and Tasks

Decisions:

- Finish Final CATME evaluations
- Finish the Design Showcase Flyer

Appendix F. Team Working Agreement

Spring 2026

1/22/2026

1. Team Information

Course Section #7440

Team Designation: N

Elizabeth Jergens

Sneha Parthe

Contact Information:

Name	Email Address	Phone Number
Nicholas Farley	Farley.322@osu.edu	908-499-8158
Riya Shankar	shankar.157@osu.edu	510-309-8215
Jonathan Yang	Yang.8041@osu.edu	765-767-3296
Dawson McMurry	mcmurry.25@osu.edu	614-598-8588

2. Team Values & Goal

What are the team's top values?

- **Communication**
- **Responsibility**
- **Considerateness**

What are the team's expectations of quality level? Top goals? Minimum acceptable goals?

Include at least 1 goal regarding psychological safety, belonging, and inclusion.

- **Our expectations for quality level is high, as we expect to put out the best work that we can as a team, our top goal being quality work in minimal time with our goal of turning assignments in on time to not back ourselves up. However, through it all, we want to prioritize respect and considerateness to each other, as a comfortable team is a cohesive team.**

3. Communication and Meetings

What are your team's preferred method(s) of contact and expected response time(s)?

The team's preferred method of contact is through the messaging group chat, and the expected response time (especially if one is directly addressed) is two hours.

How often do you plan on meeting to achieve your goals? (Do you anticipate this changing throughout the semester?)

We plan on meeting once a week and having more meetings if necessary if more discussions need to occur within the team.

Primary and Secondary Meeting Day/Time/Location

Sunday 1/2pm, Thompson Library/RPAC

Individual(s) in charge of agendas, reminders, minutes

- **Riya and Jonathan oversee reminders, Dawson is in charge of agendas, and Nick is in charge of minutes.**

4. General Expectations and Group Norms

How are team members expected to behave? What are the group norms?

- **Team members are expected to treat others with respect and with the intention of listening before speaking their opinion on a topic.**

What are acceptable/unacceptable types of interaction?

- **Acceptable interactions are those that show respect to one another and are formed in critical thinking and good intention. An unacceptable interaction is one that harms an individual mentally or physically, creating issues than resolutions.**

What are team member expectations regarding attendance?

- **Team members are expected to attend all meetings and classes unless external circumstances prevent it.**

How are team members expected to behave during lab/class periods?

- **Team members are expected to behave in a mature and positive manner throughout the labs and class, ensuring a safe and friendly collaborative environment.**

What are team members meant to do between classes? Lab/class preparation?

- **Team members are expected to complete all lab/class preparation if required, but even if not, mandatory it is in good practice to get all assignments done early.**

How are team members meant to ensure the team stays on track?

- **Staying on track means to get assignments done on time and in some cases early, ensuring that we are caught up to the task at hand.**

How are documents expected to be shared?

- **OneDrive**

How many days before an assignment is due should everybody have their portion completed for review?

- **1 Day**

When should team members first notify the group if they are struggling?

- **Notify the team members immediately. When they are struggling, they should immediately contact fellow team members for support, and if an issue is out of the member's hands, they can contact a faculty member or instructor for further help.**

5. Individual Team Member Responsibilities

When/how will individual tasks/responsibilities be assigned?

- **Individual tasks and responsibilities will be assigned through team communication. Team members can discuss and decide on the tasks/responsibilities they want to complete/undertake through group chat or before/after class discussions.**

How will the team ensure work is fairly divided and that everyone participates equally?

- **The team will communicate with other members to evenly split up the work, and everyone will have their own responsibilities that they will take ownership on.**

Are there specific roles that all team members have?

- **Every member of the team should be a leader in some way as when another member is struggling with their work or understanding an assignment, the team should be there for them to assist them throughout the way.**

What specific tasks are team members in charge of?

- **Team members are all responsible of completing their assigned work to the best of their ability and using all of their possible resources to ensure it will be sufficient work for the assignment.**

How often will these roles/task rotate?

- **Roles and tasks will be rotated and changed around each assignment, giving each member a chance at each position in a group.**

6. Conflict Resolution

- **Once the team goals, general member expectations, and individual team member responsibilities have been established, candid, non-threatening discussion must be held when the group or individuals are not meeting the agreed upon terms.**

When there is disagreement amongst members, how will the team make decisions?

- **The team will make decisions through a majority vote and active discussion about the topic of disagreement until a consensus is reached.**

How will team members above be held accountable (be specific!)?

- **Team members will be held accountable through direct addressing in the group chat or during in person meetings and members will be asked about their progress and results consistently to ensure that each member is working effectively.**

How will team members that are not meeting expectations (not contributing to the team effectively) be addressed?

- **Team members that are not meeting expectations will be addressed in the group chat or when the team meets in person. Team members will not rush to accuse or judge and will begin simply by asking the addressed members about their current situation and whether they require help and finally discuss a resolution to the conflict.**

How will team members that are not interacting appropriately with team members be addressed?

- **Team members that are not interacting appropriately with other team members will be addressed first with empathy and respect to resolve any concerns. However, if the team member refuses to cooperate with the team, the team members will reach out to the instructional team for support.**

How will the team handle resolving issues when a team member is not acting inclusively?

- **When a team member is not acting inclusively, the team will handle resolving issues by first discussing possible confrontation plans together, then approach the team member that is not acting inclusively in a polite and diplomatic manner to understand the situation at hand and work together to come up with a plan to solve the issue. If this does not work, the team can ask for help from the instructional team.**

What are the consequences for violating this agreement (be creative!)?

- **The consequences for violating this agreement will be a notification sent to the instructor and/or TAs describing the conflict, how the team intended to approach and solve it, and the results of any discussions made between the team members and the member(s) they are in conflict with. The team will allow the instructional team to come up with potential consequences for the member's actions.**

When is it okay to redefine goals, expectations, and responsibilities?

- **When the group starts to fall behind or work less efficiently, then goals and plans can be redefined. Doing so prevents group and grading conflicts as the group works with newly defined goals.**

When will UTAs, GTAs, or the instructor become involved?

- **UTAs, GTAs, or the instructor will get involved if an issue or question arises that cannot be resolved amongst the group. This ensures group cohesion and demonstrates proper resolution of any outstanding conflict.**

7. Expectations of Faculty and GTAs

- If a team member fails to live up to this agreement, the situation may be reported to the staff, but the team will still be responsible for submitting a completed assignment. Staff will be available to meet with teams to resolve issues.

8. Team Signatures

Add a screenshot of your signature or digitally sign indicating that you have read and agree to all that is contained in this agreement.



Nicholas Farley



Jonathan Yang



Riya Shankar



Dawson McMurry

Appendix G. Generative AI Prompts

Gemini – “Can you create a graphical persona, create a fictitious name in which it is a relatively elderly woman who has a hobby in gardening; however, she isn't able to do the gardening well due to mobility and joint issues/pain. The race, education, occupation, doesn't matter for this scenario. The two main tasks that she does and has pains for are, 1. Weeding and Planting - Pains: Requires constant, forceful, use of hands and forearms. Often requires resting on the arms and elbows. It is difficult to reach the roots of the weeds buried in the soil. It is often necessary to take many breaks and take pain relief medication. Task 2: Digging and Turning Soil - Pains: Using shovels can put severe strain on the arms and joints. It can cause tendonitis & carpal tunnel syndrome from gripping tools tightly. Can strain the rotor cuff due to repetitive actions.”